

Memory Effects, Dissipation, and Current Density in Time-Dependent Density Functional Theory

A Derivation-Based Graduate Tutorial

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Abstract

This tutorial addresses four questions that arise naturally when studying RT-TDDFT from a quantum-chemistry perspective: (i) what does the adiabatic approximation discard and why finite-order time-derivative corrections are insufficient; (ii) why a closed unitary quantum system can still exhibit dephasing in the one-body density matrix; (iii) why double excitations are absent from adiabatic TDDFT and what is actually required to recover them; (iv) why the charge density alone is insufficient as a basic variable for periodic systems and how current-density TDDFT resolves this. Each topic is developed through explicit derivations at the level of a first-year graduate student in quantum chemistry or condensed-matter physics.

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1 Background: The Exact TDDFT Framework

We work in atomic units ($\hbar = m_e = e = 1$) throughout. The time-dependent Kohn-Sham (TDKS) equations are:

$$i\frac{\partial\psi_n}{\partial t} = H_{\text{KS}}[\rho] \psi_n, \quad H_{\text{KS}} = -\frac{1}{2}\nabla^2 + V_{\text{ext}} + V_H[\rho] + V_{\text{xc}}[\rho], \quad (1)$$

with electron density $\rho(\mathbf{r}, t) = \sum_n f_n |\psi_n|^2$. The **occupation numbers** $f_n \in \{0, 1\}$ (or $\{0, 1, 2\}$ for spin-restricted closed-shell) are set by the **Aufbau principle** at $t = 0$: orbitals are filled from lowest to highest KS eigenvalue ϵ_n , giving $f_n = 1$ for the N lowest and $f_n = 0$ for the rest. They are *fixed at their initial values* for all $t > 0$: the TDKS equations propagate the orbital shapes $\psi_n(\mathbf{r}, t)$ but do not change which orbitals are occupied. This is why adiabatic TDDFT cannot describe T_1 population relaxation (Section 4) — the f_n are frozen constants, not dynamical variables. (Koopmans’ theorem connects ϵ_n to ionisation energies; Janak’s theorem generalises this to fractional occupations, relevant for smearing in metallic systems, but not used in the present tutorial.)

The Runge-Gross theorem. Recall the **Hohenberg-Kohn (HK) theorem** (1964): for a ground-state system, there is a one-to-one mapping between the static external potential $V_{\text{ext}}(\mathbf{r})$ and the ground-state density $\rho_0(\mathbf{r})$. Consequently, every ground-state observable is a functional of ρ_0 alone — this is the foundation of ground-state DFT.

Runge and Gross (1984) proved the time-dependent analogue of HK: given a fixed initial state Ψ_0 , there is a one-to-one mapping between the time-dependent external potential $V_{\text{ext}}(\mathbf{r}, t)$ and the density $\rho(\mathbf{r}, t)$. This guarantees the existence of an exact time-dependent XC functional:

$$V_{\text{xc}}(\mathbf{r}, t) = V_{\text{xc}}[\rho(\mathbf{r}', t'); t' \leq t](\mathbf{r}, t). \quad (2)$$

The critical feature of Eq. (2) is that V_{xc} at time t is a *functional of the entire density history* from the initial time to t — not just a function of the instantaneous density $\rho(\mathbf{r}, t)$.

The adiabatic approximation defined. The adiabatic approximation replaces the history-dependent functional by the ground-state XC functional evaluated at the instantaneous density:

$$\boxed{V_{\text{xc}}^{\text{ATDDFT}}(\mathbf{r}, t) = V_{\text{xc}}^{\text{gs}}[\rho(\cdot, t)](\mathbf{r})}. \quad (3)$$

At every timestep, Eq. (3) calls the same LDA/GGA routine used in a ground-state calculation, passing the instantaneous density. The functional has no knowledge that ρ is time-evolving.

2 Functional Derivatives: A Primer for Quantum Chemists

This section is a self-contained introduction to functional derivatives, written for readers comfortable with Gaussian-basis DFT but not variational calculus. Every symbol used later

in this tutorial— $V_{\text{xc}} = \delta E_{\text{xc}}/\delta\rho$, the XC kernel $f_{\text{xc}} = \delta V_{\text{xc}}/\delta\rho$, the linear-response equations—is a functional derivative. Understanding what these objects *are* makes the rest of the theory transparent.

Notation: Three Different Uses of δ

The symbol δ carries three distinct meanings in this section. They are easy to tell apart by context, but it is worth listing them explicitly before they all appear on the same page.

Symbol	Name	What it is
$\frac{\delta F}{\delta f(x)}$	functional derivative	a <i>function</i> of x ; measures sensitivity of F to f at x
$\delta(x - x_0)$	Dirac delta	a <i>distribution</i> ; zero everywhere except x_0 , unit integral
$\delta f, \delta\rho$	variation	an <i>infinitesimal perturbation</i> of the function f or density ρ

The functional derivative $\delta F/\delta f(x)$ is defined via an integral against a test function ϕ (Eq. 8 below): it is never a ratio of two δ 's. The Dirac delta $\delta(x - x_0)$ appears when that test function is chosen to be a spike at a single point (Eq. 10). A variation δf is the perturbation you add to f ; it plays the role of ϕ when no specific form is needed.

2.1 What Is a Functional?

In ordinary calculus a *function* maps a number to a number: $f : \mathbb{R} \rightarrow \mathbb{R}$, e.g. $f(x) = x^2$. A *functional* maps an entire *function* to a number: $F : \{\text{functions}\} \rightarrow \mathbb{R}$.

Object	Input	Output
Function $f(x)$	a number x	a number
Functional $F[f]$	an entire function $f(\cdot)$	a number

The square bracket $F[\cdot]$ is the conventional notation for a functional to distinguish it from a function $f(\cdot)$.

Examples you already know.

- **Norm:** $\|f\|^2 = \int |f(x)|^2 dx$ — maps a wavefunction to a real number.
- **Energy expectation value:** $E[\Psi] = \langle \Psi | H | \Psi \rangle$ — maps a many-body wavefunction to the energy.
- **Hartree energy:** $E_H[\rho] = \frac{1}{2} \iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d^3r d^3r'$ — maps the electron density to the classical electrostatic self-energy.
- **XC energy:** $E_{\text{xc}}[\rho]$ — maps ρ to a number whose exact form is unknown; approximated by LDA, PBE, etc.

2.2 The Functional Derivative: Intuition First

Analogy with ordinary derivatives. For a function $f(x)$, the derivative $f'(x_0)$ answers:

If I nudge the argument by ε at the point x_0 , how much does f change?

$$\delta f = f'(x_0) \delta x.$$

For a functional $F[f]$, the functional derivative $\frac{\delta F}{\delta f(x_0)}$ answers:

If I nudge the input function f by a tiny spike at the single point x_0 , how much does F change?

The Dirac Delta: Definition and Sifting Property

The **Dirac delta** $\delta(x - x_0)$ is not a function in the ordinary sense but a *distribution* (generalised function) defined by what it does inside an integral. Its two essential properties are:

$$\delta(x - x_0) = 0 \quad \text{for all } x \neq x_0, \quad (4)$$

$$\int_{-\infty}^{\infty} \delta(x - x_0) g(x) dx = g(x_0) \quad \text{for any continuous } g. \quad (5)$$

Equation (5) is the **sifting property**: the delta “picks out” the value of g at the single point x_0 .

Contrast with the Kronecker delta. The Kronecker delta δ_{ij} is an ordinary number (0 or 1) that acts on discrete indices. The Dirac delta $\delta(x - x_0)$ is a continuous analogue: it is zero everywhere except at one point, but it is normalised so that its integral equals 1. On a grid with spacing h , $\delta(x - x_0) \approx \delta_{i,i_0}/h$, which recovers the Kronecker delta in the discrete limit.

Intuitive picture. Think of $\delta(x - x_0)$ as the limit of a very narrow, very tall Gaussian: $\delta(x - x_0) = \lim_{\sigma \rightarrow 0} \frac{1}{\sigma\sqrt{2\pi}} \exp[-(x - x_0)^2/2\sigma^2]$. The area under the spike is always 1, but the width shrinks to zero.

The “tiny spike at x_0 ” is mathematically a Dirac delta: $f(x) \rightarrow f(x) + \varepsilon \delta(x - x_0)$. This is the infinitesimal perturbation of a function: it leaves f unchanged everywhere except at x_0 , where it adds ε .

Gradient analogy. If F depended on a finite set of numbers $(a_1, a_2, \dots, a_N)$ instead of on a continuous function $f(x)$, its sensitivity to each a_i would be the partial derivative $\partial F/\partial a_i$, and the full gradient would be $\nabla F = (\partial F/\partial a_1, \dots, \partial F/\partial a_N)$.

A functional $F[f]$ is the limit $N \rightarrow \infty$ of this construction, where $a_i \equiv f(x_i)$ are the values of f at a dense grid of points. The functional derivative $\delta F/\delta f(x)$ is the continuum analogue of the gradient component at grid point x :

$$\frac{\delta F}{\delta f(x)} \longleftrightarrow \left. \frac{\partial F}{\partial a_i} \right|_{a_i=f(x_i)}, \quad x_i \rightarrow x. \quad (6)$$

This is not merely suggestive — it is exact on a discrete grid.

Proof: functional derivative = partial derivative / grid spacing

Let the grid have N points $x_1, \dots, x_N$ with uniform spacing h , so f is represented by the vector $(a_1, \dots, a_N)$ with $a_i = f(x_i)$. A functional $F[f] = \int g(f(x)) dx$ is approximated by $F \approx \sum_k g(a_k) h$. Apply the definition (8) with $\phi(x) = \delta(x - x_i)$ (a spike at grid point x_i , which on the grid acts as a unit vector in the i -th direction):

$$\frac{\delta F}{\delta f(x_i)} = \lim_{\varepsilon \rightarrow 0} \frac{F[\mathbf{a} + \varepsilon \mathbf{e}_i] - F[\mathbf{a}]}{\varepsilon} = \frac{\partial F}{\partial a_i} = h g'(a_i). \quad (7)$$

But the definition also requires dividing by the “volume element” to obtain a density: comparing $\int (\delta F / \delta f(x)) \phi(x) dx$ with the finite sum $\sum_k (\partial F / \partial a_k) \phi_k$ shows that $\delta F / \delta f(x_i) = (\partial F / \partial a_i) / h$.

For the example $F = \sum_k a_k^2 h$: $\partial F / \partial a_i = 2a_i h$, so $\delta F / \delta f(x_i) = 2a_i h / h = 2a_i = 2f(x_i)$. This is exactly the continuum result $\delta F / \delta f(x) = 2f(x)$, confirmed at grid level.

2.3 Formal Definition

Definition: Functional Derivative

The functional derivative of $F[f]$ with respect to f at the point x_0 is defined by:

$$\int \frac{\delta F}{\delta f(x)} \phi(x) dx = \lim_{\varepsilon \rightarrow 0} \frac{F[f + \varepsilon \phi] - F[f]}{\varepsilon} \equiv \left. \frac{d}{d\varepsilon} F[f + \varepsilon \phi] \right|_{\varepsilon=0}, \quad (8)$$

for all “**test functions**” ϕ . A test function is any well-behaved function used to *probe* the sensitivity of F to changes in f : it represents the direction in function space along which f is perturbed. Any choice of ϕ gives a valid directional derivative; different choices probe different directions. In a Gaussian-basis code, the discrete analogue of ϕ is a variation $\delta P_{\mu\nu}$ of one density-matrix element: perturbing $P_{\mu\nu}$ by ε probes the sensitivity of $E[\mathbf{P}]$ to that element, giving $\partial E / \partial P_{\mu\nu}$ — exactly the matrix-element version of the functional derivative. Choosing $\phi(x) = \delta(x - x_0)$ and applying the sifting property (5) to the left-hand side:

$$\int \frac{\delta F}{\delta f(x)} \delta(x - x_0) dx = \frac{\delta F}{\delta f(x_0)}, \quad (9)$$

recovers the pointwise version:

$$\frac{\delta F}{\delta f(x_0)} = \left. \frac{d}{d\varepsilon} F[f + \varepsilon \delta(\cdot - x_0)] \right|_{\varepsilon=0}. \quad (10)$$

Algorithm: How to Compute a Functional Derivative

Every functional derivative calculation in this tutorial follows the same four steps.

Step 1. Perturb. Replace $f \rightarrow f + \varepsilon \phi$ in the definition of F and expand to first

order in ε , discarding $O(\varepsilon^2)$ terms.

Step 2. Subtract and divide. Form $(F[f + \varepsilon\phi] - F[f])/\varepsilon$ and take $\varepsilon \rightarrow 0$ (equivalently, $d/d\varepsilon|_{\varepsilon=0}$). The result is a linear functional of ϕ .

Step 3. Write as a single integral. Bring the result into the form $\int(\cdots)\phi(x)dx$, using symmetry of the integration variable or Fubini's theorem (switching order of a double integral) if needed.

Step 4. Read off the coefficient. The expression multiplying $\phi(x)$ inside the integral is $\delta F/\delta f(x)$. That is the functional derivative.

This procedure defines the **Gâteaux derivative** (named after René Gâteaux, 1913) — the functional analogue of the directional derivative in ordinary calculus. It is a weaker notion than the Fréchet derivative (which requires uniform convergence over all directions), but for the smooth, well-behaved functionals that arise in DFT the two coincide. No topology or functional-analysis background is required; the four-step algorithm above is all one ever needs in practice.

2.4 Worked Example 1 — A Simple Integral Functional

Let

$$F[f] = \int f(x)^2 dx. \quad (11)$$

Derivation

Perturb $f \rightarrow f + \varepsilon\phi$:

$$\begin{aligned} F[f + \varepsilon\phi] &= \int (f(x) + \varepsilon\phi(x))^2 dx \\ &= \int f(x)^2 dx + 2\varepsilon \int f(x)\phi(x) dx + \varepsilon^2 \int \phi(x)^2 dx. \end{aligned} \quad (12)$$

Subtract $F[f]$, divide by ε , and take $\varepsilon \rightarrow 0$:

$$\left. \frac{d}{d\varepsilon} F[f + \varepsilon\phi] \right|_{\varepsilon=0} = 2 \int f(x)\phi(x) dx. \quad (13)$$

Comparing with the definition (8), the coefficient of $\phi(x)$ in the integrand is $2f(x)$, so:

$$\boxed{\frac{\delta F}{\delta f(x)} = 2f(x).} \quad (14)$$

Check with ordinary calculus. On a finite grid of N points with spacing h , $F \approx \sum_i f_i^2 h$, so $\partial F/\partial f_j = 2f_j h$. Dividing by h to get the continuum limit gives $2f(x_j)$. Consistent.

2.5 Worked Example 2 — The Hartree Energy

The Hartree (Coulomb self-interaction) energy is:

$$E_H[\rho] = \frac{1}{2} \iint \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r'. \quad (15)$$

The factor of $\frac{1}{2}$ prevents **double counting**. The integrand $\rho(\mathbf{r})\rho(\mathbf{r}')/|\mathbf{r} - \mathbf{r}'|$ counts the interaction between volume elements at $\mathbf{r}$ and $\mathbf{r}'$. Because the double integral $\iint d^3r d^3r'$ runs over all ordered pairs, each unordered pair $\{\mathbf{r}, \mathbf{r}'\}$ appears twice (once as $(\mathbf{r}, \mathbf{r}')$ and once as $(\mathbf{r}', \mathbf{r})$). The $\frac{1}{2}$ divides by 2 to correct this. This is the *nonlocal* functional: E_H depends on ρ at *all* pairs of points, not just locally.

Derivation

Perturb $\rho \rightarrow \rho + \varepsilon\phi$ and expand the product:

$$\begin{aligned} E_H[\rho + \varepsilon\phi] &= \frac{1}{2} \iint \frac{(\rho(\mathbf{r}) + \varepsilon\phi(\mathbf{r}))(\rho(\mathbf{r}') + \varepsilon\phi(\mathbf{r}'))}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' \\ &= E_H[\rho] + \frac{\varepsilon}{2} \iint \frac{\rho(\mathbf{r}') \phi(\mathbf{r})}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' + \frac{\varepsilon}{2} \iint \frac{\rho(\mathbf{r}) \phi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' + O(\varepsilon^2). \end{aligned} \quad (16)$$

Relabelling dummy variables. In the second ε -linear term, swap the integration variables $\mathbf{r} \leftrightarrow \mathbf{r}'$ (both are dummy variables of integration; $|\mathbf{r} - \mathbf{r}'| = |\mathbf{r}' - \mathbf{r}|$):

$$\frac{1}{2} \iint \frac{\rho(\mathbf{r}) \phi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' \stackrel{\mathbf{r} \leftrightarrow \mathbf{r}'}{=} \frac{1}{2} \iint \frac{\rho(\mathbf{r}') \phi(\mathbf{r})}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r'. \quad (17)$$

Both ε -linear terms are now identical; their sum gives a factor of 2 that cancels the $\frac{1}{2}$:

$$E_H[\rho + \varepsilon\phi] = E_H[\rho] + \varepsilon \iint \frac{\rho(\mathbf{r}') \phi(\mathbf{r})}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' + O(\varepsilon^2). \quad (18)$$

Fubini's theorem (switching the order of the $\mathbf{r}$ and $\mathbf{r}'$ integrations, valid here because the integrand is absolutely integrable away from the diagonal) lets us write the double integral as an iterated integral with the $\mathbf{r}'$ integration performed first:

$$\left. \frac{d}{d\varepsilon} E_H[\rho + \varepsilon\phi] \right|_{\varepsilon=0} = \int \phi(\mathbf{r}) \underbrace{\left[\int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r' \right]}_{V_H(\mathbf{r})} d^3r. \quad (19)$$

The term in brackets is exactly the Hartree potential $V_H(\mathbf{r})$. So:

$$\boxed{\frac{\delta E_H}{\delta \rho(\mathbf{r})} = V_H(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'.} \quad (20)$$

Physical meaning. $V_H(\mathbf{r})$ is the electrostatic potential at $\mathbf{r}$ due to the charge distribution ρ . The functional derivative says: if you add one electron at $\mathbf{r}_0$, the electrostatic energy increases by $V_H(\mathbf{r}_0)$. This is exactly the definition of electrostatic potential from classical physics.

Physical Insight

Connection to Gaussian-basis DFT: from $\delta E_H/\delta\rho$ to $\partial E_H/\partial P_{\mu\nu}$. In a Gaussian-basis code the electron density is written $\rho(\mathbf{r}) = \sum_{\mu\nu} P_{\mu\nu} \chi_\mu(\mathbf{r}) \chi_\nu(\mathbf{r})$, where $P_{\mu\nu}$ is the one-particle density matrix and $\{\chi_\mu\}$ are the basis functions. The Hartree energy becomes a function of the matrix elements $P_{\mu\nu}$:

$$E_H = \frac{1}{2} \sum_{\mu\nu\lambda\sigma} P_{\mu\nu} P_{\lambda\sigma} (\mu\nu|\lambda\sigma), \quad (21)$$

where $(\mu\nu|\lambda\sigma) = \iint \chi_\mu(\mathbf{r}) \chi_\nu(\mathbf{r}) |\mathbf{r} - \mathbf{r}'|^{-1} \chi_\lambda(\mathbf{r}') \chi_\sigma(\mathbf{r}') d^3r d^3r'$ is the two-electron repulsion integral (ERI), familiar from Hartree-Fock.

The analogue of $\delta E_H/\delta\rho = V_H$ is the *Coulomb matrix* $\mathbf{G}$, obtained by differentiating with respect to the density matrix:

$$G_{\mu\nu} \equiv \frac{\partial E_H}{\partial P_{\mu\nu}} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma). \quad (22)$$

This is a matrix equation ($\mathbf{G} = \mathbf{J}[\mathbf{P}]$ in standard notation). The continuous functional derivative $\delta E_H/\delta\rho(\mathbf{r}) = V_H(\mathbf{r})$ and the discrete matrix derivative $\partial E_H/\partial P_{\mu\nu} = G_{\mu\nu}$ are the same object in two different representations: $V_H(\mathbf{r}) = \sum_{\mu\nu} G_{\mu\nu} \chi_\mu(\mathbf{r}) \chi_\nu(\mathbf{r})$.

The real-space grid in RMG plays exactly the role of the Gaussian basis: the continuous functional derivative $\delta E_H/\delta\rho(\mathbf{r}_i)$ at grid point $\mathbf{r}_i$ is the grid-point analogue of $G_{\mu\nu}$.

Physical Insight

The Kohn-Sham equation is derived by requiring that the total energy $E[\rho]$ is stationary under variations $\delta\rho$ subject to the constraint $\int \rho d^3r = N$. Setting $\delta E/\delta\rho = \mu$ (the chemical potential) gives the Euler equation, from which the KS potential emerges as $V_{\text{KS}} = V_{\text{ext}} + \delta E_H/\delta\rho + \delta E_{\text{xc}}/\delta\rho$. Functional derivatives are not a formalism bolted onto DFT — they *define* the KS potential.

2.6 Worked Example 3 — The Bilinear Functional

The Hartree energy is a special case of the *bilinear functional*:

$$F[f] = \iint K(x, y) f(x) f(y) dx dy, \quad (23)$$

where $K(x, y)$ is a fixed symmetric kernel, $K(x, y) = K(y, x)$. (For Hartree: $f = \rho$, $K(\mathbf{r}, \mathbf{r}') = \frac{1}{2}|\mathbf{r} - \mathbf{r}'|^{-1}$.) This example is worth working through because its first and second

functional derivatives appear in three places later in this tutorial: the Hartree potential, the Casida coupling matrix, and hybrid XC kernels.

First functional derivative.

Derivation

Perturb $f \rightarrow f + \varepsilon\phi$:

$$\begin{aligned} F[f + \varepsilon\phi] &= \iint K(x, y) (f(x) + \varepsilon\phi(x)) (f(y) + \varepsilon\phi(y)) \, dx \, dy \\ &= F[f] + \varepsilon \iint K(x, y) \phi(x) f(y) \, dx \, dy + \varepsilon \iint K(x, y) f(x) \phi(y) \, dx \, dy + O(\varepsilon^2). \end{aligned} \quad (24)$$

Rename $x \leftrightarrow y$ in the second integral and use $K(y, x) = K(x, y)$: both integrals are equal. Taking $\varepsilon \rightarrow 0$:

$$\left. \frac{d}{d\varepsilon} F[f + \varepsilon\phi] \right|_{\varepsilon=0} = 2 \int \phi(x) \underbrace{\left[\int K(x, y) f(y) \, dy \right]}_{\equiv K*f(x)} \, dx. \quad (25)$$

Reading off the coefficient of $\phi(x)$:

$$\boxed{\frac{\delta F}{\delta f(x)} = 2 \int K(x, y) f(y) \, dy.} \quad (26)$$

Check against Hartree. With $K(\mathbf{r}, \mathbf{r}') = \frac{1}{2}|\mathbf{r} - \mathbf{r}'|^{-1}$, the factor of 2 cancels the $\frac{1}{2}$ and Eq. (26) reduces to $\delta E_H / \delta \rho(\mathbf{r}) = \int \rho(\mathbf{r}') / |\mathbf{r} - \mathbf{r}'| \, d^3r' = V_H(\mathbf{r})$, consistent with Eq. (20).

Second functional derivative. Apply Eq. (38) to $\delta F / \delta f(x) = 2 \int K(x, y) f(y) \, dy$. This is a linear functional of f , so the second derivative is simply its kernel:

Derivation

Perturb $f \rightarrow f + \varepsilon \delta(\cdot - x')$ inside $\delta F / \delta f(x) = 2 \int K(x, y) f(y) \, dy$:

$$\begin{aligned} \frac{\delta}{\delta f(x')} \left[2 \int K(x, y) f(y) \, dy \right] &= 2 \int K(x, y) \delta(y - x') \, dy \\ &= 2K(x, x'). \end{aligned} \quad (27)$$

Therefore:

$$\boxed{\frac{\delta^2 F}{\delta f(x) \delta f(x')} = 2K(x, x').} \quad (28)$$

Interpretation. The second functional derivative of a bilinear functional is (twice) the kernel itself. The kernel $K(x, x')$ directly measures the nonlocality of F :

- If $K(x, y) = c\delta(x - y)$ (local kernel), $\delta^2 F/\delta f^2 = 2c\delta(x - x')$ — a diagonal Hessian, like LDA.
- If $K(x, y) = 1/|x - y|$ (Coulomb), $\delta^2 F/\delta f^2 = 2/|x - x'|$ — fully nonlocal, like exact exchange.

Physical Insight

Connection to Casida and hybrid XC. In the Casida linear-response equations (Section 5.3), the coupling matrix element is $K_{ia,jb} = \iint \phi_i(\mathbf{r})\phi_a(\mathbf{r}) \left[\frac{1}{|\mathbf{r}-\mathbf{r}'|} + f_{xc}(\mathbf{r}, \mathbf{r}') \right] \phi_j(\mathbf{r}')\phi_b(\mathbf{r}') d^3r d^3r'$. The Coulomb part is structurally $2K(x, x')$ from Eq. (28) with $K = 1/|\mathbf{r} - \mathbf{r}'|$ and the density evaluated in MO pairs. Adding f_{xc} is the XC correction to this bilinear structure. In hybrid functionals, a fraction α of the exact exchange contributes $2\alpha K(\mathbf{r}, \mathbf{r}')$ with $K = 1/2|\mathbf{r} - \mathbf{r}'|$ — again bilinear.

2.7 The XC Potential as a Functional Derivative

The total ground-state energy in DFT is decomposed as:

$$E_{\text{tot}}[\rho] = T_s[\rho] + E_{\text{ext}}[\rho] + E_H[\rho] + E_{xc}[\rho], \quad (29)$$

where $T_s[\rho]$ is the kinetic energy of the *non-interacting* KS electrons (not the true interacting kinetic energy), $E_{\text{ext}}[\rho] = \int V_{\text{ext}}(\mathbf{r})\rho(\mathbf{r}) d^3r$ is the electron-nuclear attraction, and $E_H[\rho]$ is the Hartree energy (15).

The XC energy $E_{xc}[\rho]$ is defined as whatever is left over:

$$E_{xc}[\rho] \equiv E_{\text{tot}}^{\text{exact}}[\rho] - T_s[\rho] - E_{\text{ext}}[\rho] - E_H[\rho]. \quad (30)$$

It collects the kinetic correlation (difference between true and non-interacting kinetic energies) and all exchange-correlation effects beyond the classical Hartree repulsion. Its explicit form is unknown, but its functional derivative defines the XC potential that appears in the KS equation:

$$V_{xc}(\mathbf{r}) = \frac{\delta E_{xc}[\rho]}{\delta \rho(\mathbf{r})}. \quad (31)$$

LDA: functional derivative reduces to an ordinary derivative. In the local density approximation (LDA):

$$E_{xc}^{\text{LDA}}[\rho] = \int \varepsilon_{xc}(\rho(\mathbf{r})) \rho(\mathbf{r}) d^3r, \quad (32)$$

where $\varepsilon_{xc}(\rho)$ is the XC energy per particle of a *uniform* electron gas at density ρ . The density of the uniform gas is conveniently parametrised by the **Wigner-Seitz radius** $r_s = (3/4\pi\rho)^{1/3}$ (in bohr), the radius of the sphere containing one electron on average. For real materials $r_s \approx 2\text{--}6 a_0$ (metals: $r_s \approx 2\text{--}3$; valence electrons in atoms: $r_s \approx 3\text{--}6$). The exact

$\varepsilon_{\text{xc}}(r_s)$ is known from **quantum Monte Carlo** (Ceperley and Alder, 1980) and fitted to analytic forms: the **VWN** (Vosko-Wilk-Nusair) parameterization is standard in most codes, giving ε_{xc} as a function of r_s with both exchange ($\propto r_s^{-1}$, Dirac 1930) and correlation (QMC) contributions. Because each point $\mathbf{r}$ contributes independently (no nonlocal coupling), the functional derivative reduces to an ordinary derivative of the integrand with respect to ρ :

Derivation

Perturb $\rho \rightarrow \rho + \varepsilon\phi$:

$$\begin{aligned} E_{\text{xc}}^{\text{LDA}}[\rho + \varepsilon\phi] &= \int \varepsilon_{\text{xc}}(\rho(\mathbf{r}) + \varepsilon\phi(\mathbf{r})) (\rho(\mathbf{r}) + \varepsilon\phi(\mathbf{r})) \, \text{d}^3r \\ &= \int \left[\varepsilon_{\text{xc}}(\rho) + \varepsilon\phi \varepsilon'_{\text{xc}}(\rho) + O(\varepsilon^2) \right] [\rho + \varepsilon\phi] \, \text{d}^3r \\ &= E_{\text{xc}}^{\text{LDA}}[\rho] + \varepsilon \int \phi(\mathbf{r}) \underbrace{[\varepsilon_{\text{xc}}(\rho) + \rho \varepsilon'_{\text{xc}}(\rho)]}_{=d[\rho \varepsilon_{\text{xc}}]/d\rho} \, \text{d}^3r + O(\varepsilon^2). \end{aligned} \quad (33)$$

Reading off the coefficient of $\phi(\mathbf{r})$:

$$V_{\text{xc}}^{\text{LDA}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}^{\text{LDA}}}{\delta \rho(\mathbf{r})} = \left. \frac{\text{d}[\rho \varepsilon_{\text{xc}}(\rho)]}{\text{d}\rho} \right|_{\rho=\rho(\mathbf{r})}. \quad (34)$$

Key message. For LDA, the functional derivative is an ordinary derivative of the integrand because LDA is *local in space*: the XC energy density at $\mathbf{r}$ depends only on $\rho(\mathbf{r})$, not on $\rho(\mathbf{r}')$ at any other point. For GGAs the same procedure applies but one must integrate by parts because $E_{\text{xc}}^{\text{GGA}}$ depends on $|\nabla\rho|$ as well. Here is the essential step. A GGA energy density has the form $\varepsilon_{\text{xc}}(\rho, |\nabla\rho|)$, so:

$$E_{\text{xc}}^{\text{GGA}}[\rho] = \int \varepsilon_{\text{xc}}(\rho(\mathbf{r}), |\nabla\rho(\mathbf{r})|) \rho(\mathbf{r}) \, \text{d}^3r. \quad (35)$$

Perturbing $\rho \rightarrow \rho + \varepsilon\phi$ generates a term $\varepsilon_{\text{xc},s}(\nabla\rho \cdot \nabla\phi)/|\nabla\rho|$ (where $\varepsilon_{\text{xc},s} = \partial(\rho\varepsilon_{\text{xc}})/\partial s$, $s = |\nabla\rho|$) that contains $\nabla\phi$ rather than ϕ . To read off the coefficient of ϕ one integrates by parts, using $\int f \nabla\phi \, \text{d}^3r = -\int (\nabla f)\phi \, \text{d}^3r$ (boundary term vanishes for localised densities):

$$V_{\text{xc}}^{\text{GGA}}(\mathbf{r}) = \frac{\partial(\rho \varepsilon_{\text{xc}})}{\partial \rho} - \nabla \cdot \left[\frac{\partial(\rho \varepsilon_{\text{xc}})}{\partial |\nabla\rho|} \frac{\nabla\rho}{|\nabla\rho|} \right]. \quad (36)$$

The second term is the gradient correction absent from LDA. In practice, codes evaluate both terms on the quadrature grid via automatic differentiation or analytical GGA implementations.

Physical Insight

How $V_{\text{xc}}^{\text{LDA}}$ enters the KS matrix in a Gaussian-basis code. $V_{\text{xc}}^{\text{LDA}}(\mathbf{r}) = \text{d}[\rho\varepsilon_{\text{xc}}]/\text{d}\rho|_{\rho(\mathbf{r})}$ is a *multiplicative* (local) potential: it is just a number at each point $\mathbf{r}$,

not an integral operator. In a Gaussian basis $\{\chi_\mu\}$, it enters the KS Fock matrix via:

$$(V_{\text{xc}})_{\mu\nu} = \int \chi_\mu(\mathbf{r}) V_{\text{xc}}^{\text{LDA}}(\mathbf{r}) \chi_\nu(\mathbf{r}) d^3r \approx \sum_g w_g \chi_\mu(\mathbf{r}_g) V_{\text{xc}}^{\text{LDA}}(\rho(\mathbf{r}_g)) \chi_\nu(\mathbf{r}_g), \quad (37)$$

where $\{(\mathbf{r}_g, w_g)\}$ is a numerical quadrature grid (e.g. Becke grid in molecular codes, FFT grid in plane-wave codes). This is the computational realization of Eq. (34): the functional derivative is evaluated at each grid point as an ordinary number, then integrated against basis-function pairs.

In RMG (real-space grid), each grid point *is* a quadrature point, so the sum over g is identical to the matrix element: $(V_{\text{xc}})_{ij} = \sum_g w_g \phi_i(\mathbf{r}_g) V_{\text{xc}}(\mathbf{r}_g) \phi_j(\mathbf{r}_g)$. The two codes differ only in the basis set, not in how V_{xc} is applied.

2.8 The Second Functional Derivative and the XC Kernel

Just as ordinary second derivatives $f''(x) = d^2f/dx^2$ measure the *curvature* of f , the second functional derivative $\delta^2 F / \delta f(x) \delta f(x')$ measures the *nonlocal curvature* of a functional.

Definition. The second functional derivative is defined iteratively: differentiate $\delta F / \delta f(x)$ — which is itself a functional of f — with respect to f at a second point x' :

$$\frac{\delta^2 F}{\delta f(x) \delta f(x')} = \frac{\delta}{\delta f(x')} \left[\frac{\delta F}{\delta f(x)} \right]. \quad (38)$$

Gradient analogy. On a finite grid, the first derivative is the gradient vector $g_i = \partial F / \partial a_i$. The second functional derivative is the *Hessian matrix*:

$$\frac{\delta^2 F}{\delta f(x) \delta f(x')} \longleftrightarrow H_{ij} = \frac{\partial^2 F}{\partial a_i \partial a_j}. \quad (39)$$

The Hessian H_{ij} tells you how the gradient component at grid point i changes when you change the value at grid point j . The second functional derivative $\delta^2 F / \delta f(x) \delta f(x')$ is the continuum limit: how the “gradient” $\delta F / \delta f(x)$ changes when f is nudged at a different point x' .

Physical meaning for DFT. For the XC energy, the second functional derivative is the *XC kernel*:

$$f_{\text{xc}}^{\text{gs}}(\mathbf{r}, \mathbf{r}') = \frac{\delta^2 E_{\text{xc}}}{\delta \rho(\mathbf{r}) \delta \rho(\mathbf{r}')} = \frac{\delta V_{\text{xc}}(\mathbf{r})}{\delta \rho(\mathbf{r}')}. \quad (40)$$

Reading this equation: $f_{\text{xc}}(\mathbf{r}, \mathbf{r}')$ tells you how much the XC potential at point $\mathbf{r}$ changes when the electron density is nudged by a tiny amount at the *different* point $\mathbf{r}'$. If $\mathbf{r} \neq \mathbf{r}'$, this measures the *spatial nonlocality* of the XC interaction: an electron added at $\mathbf{r}'$ “feels” the XC environment change at $\mathbf{r}$.

Worked example: second derivative of $F[f] = \int f^2 dx$. We already know from Worked Example 1 that $\delta F/\delta f(x) = 2f(x)$. Apply the definition (38): differentiate $2f(x)$ — which is a local functional of f — with respect to f at the second point x' .

Derivation

By the composite chain rule (44) with $g(f) = 2f$ and $g'(f) = 2$:

$$\frac{\delta^2 F}{\delta f(x) \delta f(x')} = \frac{\delta}{\delta f(x')} [2f(x)] = 2\delta(x - x'). \quad (41)$$

Grid check. On a grid with spacing h , $F \approx \sum_i f_i^2 h$, so $\partial F/\partial f_i = 2f_i h$ and $\partial^2 F/\partial f_i \partial f_j = 2h \delta_{ij}$. Dividing by h^2 to convert two grid-value derivatives to continuum: $2h \delta_{ij}/h^2 = 2\delta_{ij}/h \rightarrow 2\delta(x - x')$ as $h \rightarrow 0$. Consistent.

Physical meaning of the result. The Hessian of $F[f] = \int f^2 dx$ is *diagonal* in the continuum sense: nudging f at x' only changes the gradient $\delta F/\delta f(x)$ at the *same* point $x = x'$, by the amount 2. A nonlocal functional would have off-diagonal structure: nudging at x' would affect the gradient at distant $x \neq x'$. For DFT: LDA is diagonal ($f_{xc} \propto \delta(\mathbf{r} - \mathbf{r}')$); hybrid functionals and exact exchange are off-diagonal.

Why f_{xc} Controls Linear Response

Suppose the system is in its ground state and is then perturbed by $\delta V_{\text{ext}}(\mathbf{r}, \omega)$. The induced density change is:

$$\delta \rho(\mathbf{r}, \omega) = \int \chi(\mathbf{r}, \mathbf{r}', \omega) \delta V_{\text{ext}}(\mathbf{r}', \omega) d^3 r', \quad (42)$$

where χ is the interacting density-response function.

The key idea is that the KS electrons respond to the *total* effective potential $V_{\text{eff}} = V_{\text{ext}} + V_H + V_{xc}$, not just V_{ext} alone. Because V_H and V_{xc} themselves depend on $\delta \rho$, the perturbation is self-consistently amplified. Writing $\delta V_{\text{eff}} = \delta V_{\text{ext}} + (1/|\mathbf{r} - \mathbf{r}'| + f_{xc})\delta \rho$ and using $\delta \rho = \chi_s \delta V_{\text{eff}}$ (KS electrons respond to V_{eff}) gives a self-consistent equation for $\delta \rho$. Substituting $\chi \delta V_{\text{ext}} \equiv \delta \rho$ to eliminate δV_{ext} yields the Dyson equation for the interacting response χ in terms of the non-interacting KS response χ_s :

$$\chi(\mathbf{r}, \mathbf{r}', \omega) = \chi_s(\mathbf{r}, \mathbf{r}', \omega) + \iint \chi_s(\mathbf{r}, \mathbf{r}'', \omega) \left[\frac{1}{|\mathbf{r}'' - \mathbf{r}'''}| + f_{xc}(\mathbf{r}'', \mathbf{r}''', \omega) \right] \chi(\mathbf{r}''', \mathbf{r}', \omega) d^3 r'' d^3 r'''. \quad (43)$$

The XC kernel f_{xc} enters as the *exchange-correlation contribution to the effective screening*. In the adiabatic approximation, $f_{xc}(\omega) = f_{xc}^{\text{gs}}$ is frequency-independent.

Composite Chain Rule for Local Functionals

Let $g(\rho)$ be an ordinary differentiable function of a real variable, and define the functional $G[\rho]$ by pointwise evaluation: $G[\rho](\mathbf{r}) \equiv g(\rho(\mathbf{r}))$. Then:

$$\frac{\delta G[\rho](\mathbf{r})}{\delta \rho(\mathbf{r}')} = g'(\rho(\mathbf{r})) \delta(\mathbf{r} - \mathbf{r}'). \quad (44)$$

Proof

Perturb $\rho \rightarrow \rho + \varepsilon \delta(\cdot - \mathbf{r}')$ (a spike at $\mathbf{r}'$):

$$G[\rho + \varepsilon \delta(\cdot - \mathbf{r}')](\mathbf{r}) = g(\rho(\mathbf{r}) + \varepsilon \delta(\mathbf{r} - \mathbf{r}')). \quad (45)$$

Taylor-expand g to first order in ε :

$$= g(\rho(\mathbf{r})) + \varepsilon g'(\rho(\mathbf{r})) \delta(\mathbf{r} - \mathbf{r}') + O(\varepsilon^2). \quad (46)$$

Subtract $G[\rho](\mathbf{r})$, divide by ε , take $\varepsilon \rightarrow 0$:

$$\frac{\delta G[\rho](\mathbf{r})}{\delta \rho(\mathbf{r}')} = g'(\rho(\mathbf{r})) \delta(\mathbf{r} - \mathbf{r}'). \quad \square \quad (47)$$

Interpretation. Because g acts locally (it only sees $\rho(\mathbf{r})$, not ρ at any other point), nudging the density at $\mathbf{r}'$ can only affect $G(\mathbf{r})$ when $\mathbf{r}' = \mathbf{r}$. The Dirac delta enforces this: the sensitivity is $g'(\rho(\mathbf{r}))$ at $\mathbf{r}' = \mathbf{r}$ and zero everywhere else. This is precisely why LDA produces a spatially local kernel.

Computing f_{xc} for LDA. Since $V_{\text{xc}}^{\text{LDA}}(\mathbf{r}) = v_{\text{xc}}(\rho(\mathbf{r}))$ is a local function of ρ , Eq. (44) applies with $g = v_{\text{xc}}$:

Derivation

$$\begin{aligned} f_{\text{xc}}^{\text{LDA}}(\mathbf{r}, \mathbf{r}') &= \frac{\delta V_{\text{xc}}^{\text{LDA}}(\mathbf{r})}{\delta \rho(\mathbf{r}')} \\ &= \frac{\delta}{\delta \rho(\mathbf{r}')} \left[\frac{d[\rho \varepsilon_{\text{xc}}(\rho)]}{d\rho} \Big|_{\rho(\mathbf{r})} \right] \\ &\stackrel{(44)}{=} \frac{d^2[\rho \varepsilon_{\text{xc}}(\rho)]}{d\rho^2} \Big|_{\rho(\mathbf{r})} \cdot \delta(\mathbf{r} - \mathbf{r}'). \end{aligned} \quad (48)$$

The middle line identifies $g = d[\rho \varepsilon_{\text{xc}}]/d\rho$, an ordinary function of ρ ; the chain rule then supplies the factor $g' = d^2[\rho \varepsilon_{\text{xc}}]/d\rho^2$ multiplied by $\delta(\mathbf{r} - \mathbf{r}')$.

The delta function $\delta(\mathbf{r} - \mathbf{r}')$ encodes the fact that LDA is *spatially local*: nudging the density at $\mathbf{r}'$ only changes V_{xc} at that same point $\mathbf{r}' = \mathbf{r}$. A nonlocal functional (e.g. a hybrid with

exact exchange) would produce a kernel that is nonzero for $\mathbf{r} \neq \mathbf{r}'$.

2.9 Summary Table: Analogy Between Ordinary and Functional Calculus

Ordinary calculus		Functional calculus
Function $f(x) : \mathbb{R} \rightarrow \mathbb{R}$	$\longleftrightarrow$	Functional $F[f] : \{\text{functions}\} \rightarrow \mathbb{R}$
Derivative $f'(x_0)$	$\longleftrightarrow$	Functional deriv. $\delta F / \delta f(x_0)$
Gradient $\partial F / \partial a_i$	$\longleftrightarrow$	$\delta F / \delta f(x)$ at continuum point x
Hessian $\partial^2 F / \partial a_i \partial a_j$	$\longleftrightarrow$	$\delta^2 F / \delta f(x) \delta f(x')$
Chain rule: $dF/dx = \sum_i (\partial F / \partial a_i)(da_i/dx)$	$\longleftrightarrow$	$\delta F / \delta g(y) = \int (\delta F / \delta f(x))(\delta f(x) / \delta g(y)) dx$
Local function: $F(a_i)$ depends only on a_i	$\longleftrightarrow$	Local functional: $F[f]$ at x depends only on $f(x)$ (
$\partial^2 F / \partial a_i \partial a_j = 0$ for $i \neq j$ (diagonal Hessian)	$\longleftrightarrow$	$f_{\text{xc}}^{\text{LDA}}(\mathbf{r}, \mathbf{r}') \propto \delta(\mathbf{r} - \mathbf{r}')$

The functional chain rule: when does it arise? The chain rule row in the table applies whenever one functional depends on another. f is a functional of g when $f[g]$ maps functions g to functions f : each value $f(x)$ is determined by the entire function $g(\cdot)$, not just g at a single point.

When this arises in DFT. The density ρ is a functional of the external potential V_{ext} (via the ground-state KS equations), so any functional of ρ is implicitly a functional of V_{ext} . The chain rule then relates $\delta F / \delta V_{\text{ext}}$ to $\delta F / \delta \rho$ and $\delta \rho / \delta V_{\text{ext}} \equiv \chi_s$ (the KS response function).

Concrete example. Let $f(x) = \int K(x, y) g(y) dy$ (a linear transform of g) and $F[f] = \int f(x)^2 dx$. We know $\delta F / \delta f(x) = 2f(x)$ and $\delta f(x) / \delta g(y) = K(x, y)$. The chain rule gives:

$$\frac{\delta F}{\delta g(y)} = \int \frac{\delta F}{\delta f(x)} \frac{\delta f(x)}{\delta g(y)} dx = \int 2f(x) K(x, y) dx = 2 \int K(x, y) f(x) dx. \quad (49)$$

Check by direct calculation. $F[g] = \int (\int K(x, y) g(y) dy)^2 dx$. Perturb $g \rightarrow g + \varepsilon \phi$ and take $\varepsilon \rightarrow 0$: $\delta F / \delta g(y) = 2 \int K(x, y) f(x) dx$. Consistent. ✓

Physical Insight

Why $f_{\text{xc}}^{\text{LDA}}$ is a delta function in space but not in time (for TDDFT). Spatially, LDA is local: $V_{\text{xc}}(\mathbf{r})$ depends only on $\rho(\mathbf{r})$, not on $\rho(\mathbf{r}')$ at other points. This makes $\delta V_{\text{xc}}(\mathbf{r}) / \delta \rho(\mathbf{r}') \propto \delta(\mathbf{r} - \mathbf{r}')$. The adiabatic approximation further makes it a delta function *in time*: $f_{\text{xc}}(\mathbf{r}, \mathbf{r}', t - t') = f_{\text{xc}}^{\text{gs}}(\mathbf{r}, \mathbf{r}') \delta(t - t')$. Both delta functions together say: the XC response has zero range in space and zero memory in time. This is what Section 3 now analyzes in detail.

3 Memory: What Is Discarded and Why Derivatives Are Insufficient

3.1 Memory as a Convolution Integral

Functional Taylor Expansion

Any sufficiently smooth functional $F[f]$ can be expanded around a reference f_0 in powers of the deviation $\delta f = f - f_0$:

$$F[f_0 + \delta f] = F[f_0] + \int \frac{\delta F}{\delta f(x)} \Big|_{f_0} \delta f(x) dx + \frac{1}{2} \iint \frac{\delta^2 F}{\delta f(x) \delta f(x')} \Big|_{f_0} \delta f(x) \delta f(x') dx dx' + \dots \quad (50)$$

This is the functional analogue of the ordinary Taylor series $F(x_0 + \delta x) = F(x_0) + F'(x_0)\delta x + \frac{1}{2}F''(x_0)(\delta x)^2 + \dots$, with the sum over discrete terms replaced by integrals over continuous variables. Truncating at first order gives the **linear response** approximation:

$$\delta F \equiv F[f_0 + \delta f] - F[f_0] \approx \int \frac{\delta F}{\delta f(x)} \Big|_{f_0} \delta f(x) dx. \quad (51)$$

To see concretely what “memory” means, linearize V_{xc} around a reference density ρ_0 using Eq. (51). For a small perturbation $\delta\rho(\mathbf{r}, t)$:

$$\delta V_{xc}(\mathbf{r}, t) = \int_0^t dt' \int d^3r' \underbrace{\frac{\delta V_{xc}(\mathbf{r}, t)}{\delta \rho(\mathbf{r}', t')}}_{f_{xc}(\mathbf{r}, \mathbf{r}', t - t')} \delta \rho(\mathbf{r}', t'). \quad (52)$$

The quantity $f_{xc}(\mathbf{r}, \mathbf{r}', \tau)$ is the *XC kernel*. **It is exactly the second functional derivative of the XC energy computed in Section 2**, now promoted to a time-dependent object:

$$f_{xc}(\mathbf{r}, \mathbf{r}', t - t') = \frac{\delta V_{xc}(\mathbf{r}, t)}{\delta \rho(\mathbf{r}', t')} = \frac{\delta^2 E_{xc}[\rho]}{\delta \rho(\mathbf{r}, t) \delta \rho(\mathbf{r}', t')}. \quad (53)$$

In the ground-state (static) limit $t \rightarrow t'$, this reduces to $f_{xc}^{gs}(\mathbf{r}, \mathbf{r}') = \delta^2 E_{xc} / \delta \rho(\mathbf{r}) \delta \rho(\mathbf{r}')$ from Eq. (40). The memory structure ($\tau = t - t'$ dependence) is the new feature: it encodes how a density perturbation at time t' continues to affect the XC potential at all later times $t > t'$, i.e., XC has a causal memory kernel. $f_{xc} = 0$ for $\tau < 0$ (causality).

Physical Insight

The exact V_{xc} is a **convolution over history**: the XC response today depends on the whole trajectory $\delta\rho(\mathbf{r}', t')$ for all past times $t' \leq t$, weighted by the memory kernel $f_{xc}(\tau)$. This is analogous to a viscoelastic material whose stress today reflects the full history of its deformation — not just its current state.

The adiabatic approximation drops the time-nonlocality entirely. In the adiabatic approximation, $V_{\text{xc}}(\mathbf{r}, t) = V_{\text{xc}}^{\text{gs}}[\rho(\cdot, t)](\mathbf{r})$ depends only on the density *at the same instant* t , not on any earlier time t' . Therefore, varying with respect to $\rho(\mathbf{r}', t')$:

- For $t' = t$: the variation gives the *static* XC kernel $f_{\text{xc}}^{\text{gs}}(\mathbf{r}, \mathbf{r}') = \delta V_{\text{xc}}^{\text{gs}} / \delta \rho(\mathbf{r}')$, exactly the ground-state second functional derivative.
- For $t' \neq t$: $V_{\text{xc}}(t)$ does not depend on $\rho(t')$ at all, so $\delta V_{\text{xc}}(\mathbf{r}, t) / \delta \rho(\mathbf{r}', t') = 0$.

Combined, the adiabatic kernel is:

$$f_{\text{xc}}^{\text{ATDDFT}}(\mathbf{r}, \mathbf{r}', \tau) = f_{\text{xc}}^{\text{gs}}(\mathbf{r}, \mathbf{r}') \delta(\tau), \quad (54)$$

a delta function in time: the response depends only on the density at the same instant, with zero memory.

3.2 The Frequency-Domain Picture

Causality — $f_{\text{xc}}(\tau) = 0$ for $\tau < 0$ — means the standard two-sided Fourier transform $\int_{-\infty}^{\infty} f_{\text{xc}}(\tau) e^{i\omega\tau} d\tau$ reduces to a one-sided integral: the $\tau < 0$ half contributes nothing. This one-sided transform is the **Laplace transform evaluated on the real axis** (or equivalently, the Fourier transform restricted to causal functions):

$$f_{\text{xc}}(\mathbf{r}, \mathbf{r}', \omega) = \int_0^{\infty} f_{\text{xc}}(\mathbf{r}, \mathbf{r}', \tau) e^{i\omega\tau} d\tau. \quad (55)$$

The sign convention $e^{+i\omega\tau}$ (rather than $e^{-i\omega\tau}$) is chosen so that $f_{\text{xc}}(\omega)$ is analytic in the upper complex half-plane $\text{Im } \omega > 0$, which is the standard physics convention for retarded response functions and is required for the Kramers-Kronig relations (Section ??). The adiabatic kernel corresponds to:

$$f_{\text{xc}}^{\text{ATDDFT}}(\omega) = f_{\text{xc}}^{\text{gs}} = \text{const.} \quad (\text{independent of } \omega). \quad (56)$$

The *exact* $f_{\text{xc}}(\omega)$ has a much richer structure.

Why response functions have poles: the resonance picture. Consider driving a quantum system with a perturbation at frequency ω . The response $\delta\rho(\omega)$ is proportional to the resolvent $(H - \omega)^{-1}$, which diverges whenever ω equals an eigenvalue of H . In Dirac notation, the exact density-response function has the **Lehmann representation**:

$$\chi(\mathbf{r}, \mathbf{r}', \omega) = \sum_k \left[\frac{\langle 0 | \hat{\rho}(\mathbf{r}) | k \rangle \langle k | \hat{\rho}(\mathbf{r}') | 0 \rangle}{\omega - \Omega_k + i\eta} - \frac{\langle 0 | \hat{\rho}(\mathbf{r}') | k \rangle \langle k | \hat{\rho}(\mathbf{r}) | 0 \rangle}{\omega + \Omega_k + i\eta} \right], \quad (57)$$

where $\Omega_k = E_k - E_0 > 0$ are the exact excitation energies and $|k\rangle$ the exact excited states ($\eta \rightarrow 0^+$ enforces causality). Each term $1/(\omega - \Omega_k)$ is a *simple pole* at $\omega = \Omega_k$: a sharp divergence at the resonance frequency. Physically, near $\omega \approx \Omega_k$ the perturbation drives the system at exactly the frequency where it “wants to oscillate,” producing an unbounded linear response (bounded in practice by damping or finite pulse duration).

Because f_{xc} is defined via the Dyson equation (43) relating χ_s (KS, poles at KS excitation energies) to χ (interacting, poles at exact excitation energies Ω_k), f_{xc} must itself have poles at Ω_k to shift the KS poles to the correct positions. For a finite system:

$$f_{\text{xc}}(\omega) \sim \sum_k \frac{R_k}{\omega - \Omega_k} + \text{regular terms}, \quad (58)$$

where R_k are residues encoding the coupling strengths. The imaginary part $\text{Im } f_{\text{xc}}(\omega) \neq 0$ for $\omega > 0$ encodes dissipation.

3.3 Why Time Derivatives Cannot Replace Full Memory

A natural attempt to go beyond the adiabatic limit is to include time derivatives of the density:

$$V_{\text{xc}}[\rho](t) \approx V_{\text{xc}}^{\text{gs}}[\rho(t)] + \alpha_1 \frac{\partial \rho}{\partial t} + \alpha_2 \frac{\partial^2 \rho}{\partial t^2} + \dots \quad (59)$$

This is the *adiabatic gradient expansion in time*.

In frequency space, including terms up to order N corresponds to replacing the full $f_{\text{xc}}(\omega)$ by a polynomial of degree N :

$$f_{\text{xc}}^{\text{ATDDFT}+N}(\omega) \approx c_0 + c_1\omega + c_2\omega^2 + \dots + c_N\omega^N. \quad (60)$$

Key Result

A polynomial in ω cannot reproduce a function with poles. No finite-order derivative expansion can capture the poles in Eq. (58). The poles are essential for:

1. Exact resonance positions of excitation energies.
2. Finite linewidths (imaginary part of poles).
3. Double excitations (poles at two-electron excitation frequencies).

Derivation

Explicit demonstration. Suppose the exact $f_{\text{xc}}(\omega)$ has a single pole at Ω_1 :

$$f_{\text{xc}}(\omega) = \frac{R_1}{\omega - \Omega_1}.$$

Its Taylor expansion around $\omega = 0$ is:

$$f_{\text{xc}}(\omega) = -\frac{R_1}{\Omega_1} \sum_{n=0}^{\infty} \left(\frac{\omega}{\Omega_1} \right)^n = -\frac{R_1}{\Omega_1} - \frac{R_1}{\Omega_1^2} \omega - \frac{R_1}{\Omega_1^3} \omega^2 - \dots$$

This series *diverges* for $|\omega| > |\Omega_1|$ and never converges to the correct form at $\omega = \Omega_1$. The pole cannot be represented by any finite polynomial.

Deeper reason (complex analysis). A polynomial $p(\omega) = \sum_{n=0}^N c_n \omega^n$ is an *entire function*: it is analytic everywhere in the complex ω -plane and has no singularities at any finite point. A function with a pole at Ω_1 is *not* entire — it is meromorphic,

with a singularity at Ω_1 . No entire function can equal a meromorphic function: their analyticity structures are fundamentally different. Increasing N does not help; adding more polynomial terms keeps the function entire. The mismatch is not a convergence-radius problem but a topological one: you cannot continuously deform an analytic function into one with a pole.

Physically: near $\omega \approx \Omega_1$, the electron system has a resonant response that persists in time. The polynomial approximation has no singularity and therefore no resonant buildup.

Remark 1. Driving with $\cos(\omega_{\text{drive}} t)$ at the resonance frequency does produce a resonant buildup in the *total* density response even in adiabatic TDDFT, because the KS system has its own resonance near ω_0 . What the adiabatic approximation gets wrong are: (a) the exact position of ω_0 (determined by the poles of f_{xc}); (b) the linewidth ($\text{Im } f_{\text{xc}} = 0$ in the adiabatic limit); (c) states dominated by double-excitation character (Section 5).

4 Dissipation, Unitarity, and Dephasing

4.1 The Apparent Paradox

Adiabatic TDDFT uses a Hermitian KS Hamiltonian $H_{\text{KS}}^\dagger = H_{\text{KS}}$. The KS one-body density matrix is $P_{ij}(t) = \langle \psi_j | \psi_i \rangle$ (more precisely, for a set of occupied orbitals, $P = \sum_n f_n |\psi_n\rangle \langle \psi_n|$). Differentiating with respect to time and using the TDKS equation $i\partial_t |\psi_n\rangle = H_{\text{KS}} |\psi_n\rangle$ (and its conjugate $-i\partial_t \langle \psi_n| = \langle \psi_n| H_{\text{KS}}$):

$$i\dot{P} = i \sum_n f_n (\partial_t |\psi_n\rangle \langle \psi_n| + |\psi_n\rangle \partial_t \langle \psi_n|) = \sum_n f_n (H_{\text{KS}} |\psi_n\rangle \langle \psi_n| - |\psi_n\rangle \langle \psi_n| H_{\text{KS}}) = [H_{\text{KS}}, P]. \quad (61)$$

This is the **von Neumann equation** $i\dot{P} = [H_{\text{KS}}, P]$.

Physical Insight

MO-basis vs AO-basis density matrix: connecting RMG to Gaussian codes.

RMG propagates $P(t)$ in the basis of *ground-state KS orbitals* $\{\phi_n(\mathbf{r})\}$ (MO basis), so $P_{ij}(t) = \langle \phi_j | P(t) | \phi_i \rangle$ is a matrix in MO space.

Gaussian codes such as NWChem or Gaussian propagate the density matrix in the *atomic orbital* (AO) basis $\{\chi_\mu(\mathbf{r})\}$:

$$P_{\mu\nu}(t) = \langle \chi_\nu | P(t) | \chi_\mu \rangle = \sum_{kl} C_{\mu k}^* P_{kl}(t) C_{\nu l}, \quad (62)$$

where $C_{\mu n}$ is the MO coefficient matrix ($\phi_n = \sum_\mu C_{\mu n} \chi_\mu$). In matrix notation: $\mathbf{P}_{\text{AO}} = \mathbf{C} \mathbf{P}_{\text{MO}} \mathbf{C}^\dagger$.

The von Neumann equation in the AO basis includes the overlap matrix $S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle$ (non-orthogonal basis):

$$i\mathbf{S} \dot{\mathbf{P}}_{\text{AO}} = \mathbf{H}_{\text{AO}} \mathbf{P}_{\text{AO}} \mathbf{S} - \mathbf{S} \mathbf{P}_{\text{AO}} \mathbf{H}_{\text{AO}}. \quad (63)$$

In the MO basis $\mathbf{S} = \mathbf{1}$ (orthonormal KS orbitals), so the equation reduces to the clean form $i\dot{P} = [H, P]$ used in RMG. The two representations are physically identical; the MO basis simply removes the overhead of carrying $\mathbf{S}$.

The **solution** of the von Neumann equation over one timestep Δt is given by the **Magnus propagator**. The equation of motion $i\dot{P} = [H, P]$ is an operator differential equation; its formal solution for a time-dependent H is *not* simply $P(t) = e^{-iHt}P(0)e^{+iHt}$ (that would only be correct for constant H). Instead, the Magnus expansion provides the exact solution as $P(t + \Delta t) = e^{-i\Omega}P(t)e^{+i\Omega}$, where Ω is a Hermitian operator built from time-ordered integrals of H . To second order (the level used in RMG):

$$P(t + \Delta t) = e^{-i\Omega} P(t) e^{+i\Omega}, \quad \Omega \approx \frac{1}{2}(H(t) + H(t + \Delta t)) \Delta t, \quad \Omega = \Omega^\dagger. \quad (64)$$

This is a unitary similarity transform: $e^{-i\Omega}$ is the propagator, and its Hermitian conjugate $e^{+i\Omega}$ appears because P carries a ket on the left and a bra on the right ($P = |\psi\rangle\langle\psi|$). One might then ask: can the exact TDDFT functional (with memory) restore dissipation? And does unitarity preclude all relaxation?

The answer requires distinguishing three different things: the N -body wavefunction, the one-body density matrix, and what it means to have a “closed” system.

4.2 Unitarity of the Magnus Propagator and Conservation of Occupations

Derivation

Eigenvalues of P are conserved under unitary propagation.

Let $P(t) = V\Lambda V^\dagger$ be the eigendecomposition of P at time t , with eigenvalues $\Lambda = \text{diag}(f_1, f_2, \dots)$ (occupation numbers). Under the Magnus step Eq. (64):

$$\begin{aligned} P(t + \Delta t) &= e^{-i\Omega} V \Lambda V^\dagger e^{+i\Omega} \\ &= \underbrace{(e^{-i\Omega} V)}_{\tilde{V}} \Lambda \underbrace{(e^{+i\Omega} V^\dagger)^\dagger}_{\tilde{V}^\dagger}. \end{aligned} \quad (65)$$

Since $e^{-i\Omega}$ is unitary (because Ω is Hermitian), $\tilde{V}$ is unitary, and $P(t + \Delta t) = \tilde{V} \Lambda \tilde{V}^\dagger$ has the *same eigenvalues* Λ .

Consequence: the occupation numbers $\{f_n\}$ are constants of motion in KS-TDDFT. No population can transfer from an excited state to the ground state — ever. This is exact regardless of whether V_{xc} has memory.

Therefore: **T₁ relaxation (population decay) is absent in TDDFT**, not because of the adiabatic approximation, but because the KS Hamiltonian is Hermitian and the evolution is unitary.

4.3 T_1 and T_2 : The Bloch Equation Analogy

The language of T_1 and T_2 relaxation, familiar from NMR spectroscopy, clarifies what can and cannot be captured by TDDFT. Consider a two-level system with ground state $|g\rangle$ and excited state $|e\rangle$. The density matrix is:

$$\rho = \begin{pmatrix} \rho_{ee} & \rho_{eg} \\ \rho_{ge} & \rho_{gg} \end{pmatrix}, \quad \rho_{ee} + \rho_{gg} = 1. \quad (66)$$

The Bloch equations with relaxation read:

$$\dot{\rho}_{ee} = -\frac{\rho_{ee}}{T_1} + (\text{driving terms}), \quad (67)$$

$$\dot{\rho}_{eg} = -i\omega_0\rho_{eg} - \frac{\rho_{eg}}{T_2} + (\text{driving terms}), \quad (68)$$

where $\omega_0 = E_e - E_g$ is the transition frequency.

- T_1 (**longitudinal, population relaxation**): excited population ρ_{ee} decays exponentially. In spectroscopy this is spontaneous emission + non-radiative decay. Physical mechanism: coupling to the radiation field (QED photon bath) or phonons. *Not present in TDDFT, which describes only electrons.*
- T_2 (**transverse, dephasing**): the off-diagonal element ρ_{eg} (coherence) decays as e^{-t/T_2} . The general bound is $T_2 \leq 2T_1$. The excess decay beyond the $2T_1$ limit is called *pure dephasing* (T_2^*) and arises from elastic processes that randomize the quantum phase without changing populations:

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_2^*}. \quad (69)$$

Physical Insight

Your atom-in-vacuum example is exactly right. A single atom in perfect vacuum with no external coupling has: $T_1 = \infty$ (no radiation bath to emit into — spontaneous emission is a QED effect, absent from the Schrödinger equation for electrons alone), and $T_2 = \infty$ (no elastic collisions to dephase the coherence). The excited state is stable forever. This is the correct quantum-mechanical result in the absence of a bath.

4.4 The Reduced Density Matrix and Intrinsic Dephasing

Here is the resolution of the apparent paradox between unitary evolution and dephasing.

The exact N -electron wavefunction $\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N, t)$ evolves unitarily. The *one-body reduced density matrix* (1-RDM):

$$\gamma(\mathbf{r}, \mathbf{r}', t) = N \int \Psi^*(\mathbf{r}, \mathbf{r}_2, \dots) \Psi(\mathbf{r}', \mathbf{r}_2, \dots) d\mathbf{r}_2 \cdots d\mathbf{r}_N \quad (70)$$

is obtained by tracing out $N - 1$ electrons. Its equation of motion is:

$$i\dot{\gamma} = [T + V_{\text{ext}}, \gamma] + \mathcal{C}, \quad (71)$$

where $\mathcal{C}$ is the *collision integral* — the contribution from the two-body correlations that survive after tracing over $N - 1$ electrons. This term is *non-unitary* for the 1-RDM even though the N -body evolution is unitary. It is the source of intrinsic dephasing from electron-electron scattering.

Contrast with the KS density matrix. The KS system is *non-interacting* by construction: its N electrons move independently in the effective potential V_{KS} . Tracing over $N - 1$ KS electrons produces no residual two-body correlation because there are none — so $\mathcal{C} = 0$ for the KS system. The KS density matrix therefore satisfies the *homogeneous* equation:

$$i\dot{P} = [H_{\text{KS}}, P], \quad \mathcal{C} = 0. \quad (72)$$

Both γ and P satisfy equations that *look* like $i\dot{M} = [H, M]$, but they describe different objects at different levels of theory. The collision integral $\mathcal{C}$ is precisely what P is missing compared to γ , and it is what would be needed for physical dephasing.

Derivation

Why $\mathcal{C}$ causes pure dephasing.

Decompose $\mathcal{C}$ into a Hermitian part $\mathcal{C}_H$ and an anti-Hermitian part $\mathcal{C}_A$:

$$\mathcal{C} = \mathcal{C}_H + \mathcal{C}_A, \quad \mathcal{C}_H = \frac{1}{2}(\mathcal{C} + \mathcal{C}^\dagger), \quad \mathcal{C}_A = \frac{1}{2}(\mathcal{C} - \mathcal{C}^\dagger).$$

$\mathcal{C}_H$ renormalizes the effective Hamiltonian (energy shifts). $\mathcal{C}_A$ is non-Hermitian and causes the *magnitude* of off-diagonal elements to decay:

$$i\dot{\gamma}_{ij} = (\epsilon_i - \epsilon_j)\gamma_{ij} + \mathcal{C}_{A,ij}.$$

If $\mathcal{C}_A = -i\Gamma\gamma$ with $\Gamma > 0$, then $\gamma_{ij}(t) \propto e^{-i\omega_{ij}t}e^{-\Gamma t}$ — the coherence decays exponentially while the diagonal populations (in the eigenstate basis) remain conserved. This is pure dephasing with $T_2^* = 1/\Gamma$.

In TDDFT, the exact $f_{\text{xc}}(\omega)$ with $\text{Im } f_{\text{xc}}(\omega) \neq 0$ plays the role of $\mathcal{C}_A$. Here is why. The linearised TDKS equation for a density perturbation $\delta\rho$ reads (in frequency space, using Eq. 50):

$$\delta\rho(\omega) = \chi_s \left[\delta V_{\text{ext}} + \frac{1}{|\mathbf{r} - \mathbf{r}'|} \delta\rho + f_{\text{xc}}(\omega) \delta\rho \right](\omega). \quad (73)$$

When $f_{\text{xc}}(\omega)$ is real, the effective KS Hamiltonian that drives $\delta\rho$ is Hermitian, and the response is purely oscillatory (no decay). When $\text{Im } f_{\text{xc}}(\omega) \neq 0$, the effective driving potential acquires an imaginary part, which acts like a damping term in the equation of motion for the off-diagonal elements of P . Specifically, an off-diagonal coherence P_{nm} evolves as $i\dot{P}_{nm} \approx (\omega_{nm} - i\Gamma_{nm})P_{nm}$, where $\Gamma_{nm} \propto \text{Im } f_{\text{xc}}(\omega_{nm}) > 0$ is the dephasing rate. This is the same exponential decay produced by the collision integral $\mathcal{C}_A$ in the reduced density matrix equation: it introduces an effective non-Hermitian correction to the KS Hamiltonian in the frequency domain, providing T_2 dephasing without violating N -body unitarity.

Summary.

Relaxation	Physical origin	In TDDFT?
T_1 (population)	Coupling to radiation/phonon bath	No
T_2 pure dephasing	Electron-electron scattering ($\mathcal{C}_A$)	Only with $\text{Im } f_{xc} \neq 0$
T_2 from T_1	Same bath as T_1	No

5 Double Excitations: Selection Rules and CI Mixing

5.1 Pure Double Excitations Are Dipole-Forbidden — Your Intuition Is Correct

Consider two non-interacting helium atoms A and B, separated in space. Define:

$$\begin{aligned}
 |G_A\rangle &= \text{He}_A \text{ ground state } (1s)^2, \\
 |E_A\rangle &= \text{He}_A \text{ singly excited state } (1s)(2p), \\
 |G_B\rangle, |E_B\rangle &= \text{same for He}_B.
 \end{aligned}$$

The composite two-atom states are tensor products:

$$\begin{aligned}
 |\Psi_0\rangle &= |G_A\rangle \otimes |G_B\rangle \equiv |GG\rangle, \\
 |\Psi_S^A\rangle &= |E_A\rangle \otimes |G_B\rangle \equiv |EG\rangle, \\
 |\Psi_S^B\rangle &= |G_A\rangle \otimes |E_B\rangle \equiv |GE\rangle, \\
 |\Psi_D\rangle &= |E_A\rangle \otimes |E_B\rangle \equiv |EE\rangle.
 \end{aligned}$$

The dipole operator for the whole system decomposes as:

$$\hat{x} = \hat{x}_A \otimes \hat{1}_B + \hat{1}_A \otimes \hat{x}_B. \quad (74)$$

Derivation

Transition moment from ground to doubly excited state.

$$\begin{aligned}
 \langle EE|\hat{x}|GG\rangle &= \langle EE|\hat{x}_A \otimes \hat{1}_B + \hat{1}_A \otimes \hat{x}_B|GG\rangle \\
 &= \underbrace{\langle E_A|\hat{x}_A|G_A\rangle}_{\neq 0} \underbrace{\langle E_B|G_B\rangle}_{=0} + \underbrace{\langle E_A|G_A\rangle}_{=0} \underbrace{\langle E_B|\hat{x}_B|G_B\rangle}_{\neq 0} \\
 &= 0.
 \end{aligned} \quad (75)$$

Both terms vanish because excited and ground states on the *other* atom are orthogonal. The dipole operator connects only states that differ by a *single* local excitation; reaching $|EE\rangle$ requires both atoms to be excited simultaneously, which a one-body operator cannot achieve.

Physical Insight

You are correct: a pure doubly excited state is dipole-dark from the ground state. To reach $|EE\rangle$ directly requires a two-photon operator $\hat{x}_A\hat{x}_B$ — which is accessible in two-photon absorption spectroscopy. This is a different (and real) phenomenon. The TDDFT “double excitation problem” concerns a *different* situation: states that are **predominantly** doubly excited but have **mixed** character due to Coulomb coupling, and that *are* dipole-accessible.

5.2 Turning On the Coulomb Interaction: CI Mixing

Now allow a Coulomb interaction between the two He atoms:

$$V_{ee}^{AB} = \sum_{i \in A} \sum_{j \in B} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}. \quad (76)$$

Treat V_{ee}^{AB} as a perturbation. By first-order degenerate perturbation theory, the eigenstates of $H_0 + V_{ee}^{AB}$ are linear combinations of the unperturbed states. In the subspace $\{|EG\rangle, |GE\rangle, |EE\rangle, \dots\}$:

Derivation

First-order CI mixing of $|EE\rangle$ into $|EG\rangle$.

The coupling matrix element between $|EG\rangle$ and $|EE\rangle$ is:

$$\langle EG | V_{ee}^{AB} | EE \rangle = \langle E_A, G_B | V_{ee}^{AB} | E_A, E_B \rangle.$$

Since V_{ee}^{AB} involves coordinates of both atoms, this is generically nonzero — it is the inter-atomic Coulomb integral.

To first order in V_{ee}^{AB} , the corrected eigenstate is:

$$|\tilde{\Psi}_1\rangle = \alpha|EG\rangle + \beta|GE\rangle + \underbrace{\frac{\langle EE | V_{ee}^{AB} | EG \rangle}{E_{EG} - E_{EE}}}_{\gamma} |EE\rangle + \dots \quad (77)$$

The coefficient γ is generically nonzero; the state $|\tilde{\Psi}_1\rangle$ has genuine double-excitation character.

Now compute the dipole transition moment from the ground state to $|\tilde{\Psi}_1\rangle$:

$$\begin{aligned} \langle \tilde{\Psi}_1 | \hat{x} | GG \rangle &= \alpha \underbrace{\langle EG | \hat{x} | GG \rangle}_{\neq 0} + \beta \underbrace{\langle GE | \hat{x} | GG \rangle}_{\neq 0} + \gamma \underbrace{\langle EE | \hat{x} | GG \rangle}_{=0} \\ &= \alpha \langle E_A | \hat{x}_A | G_A \rangle + \beta \langle E_B | \hat{x}_B | G_B \rangle \neq 0. \end{aligned} \quad (78)$$

Key Result

Once the Coulomb interaction mixes a double-excitation component into an otherwise singly-excited state, that state becomes:

1. **Dipole-accessible** from the ground state (nonzero transition moment), and
2. **At an energy shifted** by the CI mixing — an energy that adiabatic TDDFT cannot reproduce.

The TDDFT failure is primarily **energetic**: the state appears in the spectrum, but at the wrong position.

5.3 The Casida Linear-Response Equations

Linear-response TDDFT (Casida 1995) reframes excitation energies as eigenvalues of a matrix equation. Starting from the linear response of the TDKS density to an external perturbation $V_{\text{ext}}(\omega)$, one derives the response function:

$$\delta\rho(\omega) = \chi(\omega) V_{\text{ext}}(\omega), \quad \chi(\omega) = \chi_0(\omega) [1 - (v_{ee} + f_{xc}(\omega))\chi_0(\omega)]^{-1}, \quad (79)$$

where χ_0 is the non-interacting KS response function and $v_{ee}(\mathbf{r}, \mathbf{r}') = 1/|\mathbf{r} - \mathbf{r}'|$.

Poles of $\chi(\omega)$ are the excitation frequencies. In the molecular orbital (MO) basis $\{\phi_i, \phi_a\}$ (occupied i , virtual a), expanding χ_0 in terms of KS orbital pairs gives the Casida eigenvalue problem:

$$\begin{pmatrix} A(\omega) & B \\ B^* & A^*(\omega) \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \omega \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}, \quad (80)$$

where:

$$A_{ia,jb}(\omega) = (\epsilon_a - \epsilon_i)\delta_{ij}\delta_{ab} + K_{ia,jb}(\omega), \quad (81)$$

$$B_{ia,jb} = K_{ia,bj}, \quad (82)$$

$$K_{ia,jb}(\omega) = \iint \phi_i^*(\mathbf{r})\phi_a(\mathbf{r}) \left[\frac{1}{|\mathbf{r} - \mathbf{r}'|} + f_{xc}(\mathbf{r}, \mathbf{r}', \omega) \right] \phi_j(\mathbf{r}')\phi_b^*(\mathbf{r}') d^3r d^3r'. \quad (83)$$

Physical Insight

Connection to Gaussian-basis CI/RPA: $K_{ia,jb}$ and the ERI $(ia|jb)$. The Coulomb part of $K_{ia,jb}$ is exactly a two-electron repulsion integral (ERI) in the MO basis:

$$K_{ia,jb}^{\text{Coulomb}} = \iint \frac{\phi_i^*(\mathbf{r})\phi_a(\mathbf{r})\phi_j(\mathbf{r}')\phi_b^*(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' \equiv (ia|jb). \quad (84)$$

In Gaussian-basis codes this integral is stored in the two-electron integral file and is familiar from CIS, RPA, and TDDFT calculations. The full Casida matrix is therefore:

$$K_{ia,jb}(\omega) = (ia|jb) + (ia|f_{xc}|jb), \quad (85)$$

where $(ia|f_{xc}|jb) = \iint \phi_i^*\phi_a f_{xc} \phi_j\phi_b^* d^3r d^3r'$ is the XC correction.

Method	$K_{ia,jb}$	Notes
CIS (HF)	$(ia jb) - (ij ab)$	Coulomb + exchange, no correlation
RPA (adiabatic)	$(ia jb)$	Coulomb only (Tamm-Dancoff approx.)
TDDFT-LDA	$(ia jb) + (ia f_{xc}^{LDA} jb)$	Coulomb + local XC
Exact TDDFT	$(ia jb) + (ia f_{xc}(\omega) jb)$	frequency-dep. XC kernel

The real-space grid in RMG evaluates the same integrals numerically: $\sum_i \phi_i^*(\mathbf{r}_i) \phi_a(\mathbf{r}_i) V_H(\mathbf{r}_i) h^3$ is the grid-point analogue of $(ia|jb)$ contracted over the Hartree potential.

Dimension of the Casida matrix. The indices (i, a) and (j, b) run over *single* KS excitation pairs: $N_{\text{occ}} \times N_{\text{virt}}$ in total. **Double excitations** $(i, j) \rightarrow (a, b)$ **are not included as basis vectors.**

5.4 Why Double Excitations Require Poles in $f_{xc}(\omega)$

To see why a pole in f_{xc} is needed, consider what happens when a doubly excited state $|D\rangle$ at energy Ω_D is near a singly excited KS pair $(i \rightarrow a)$ at energy $\omega_{ia} = \epsilon_a - \epsilon_i$.

Derivation

A pole in f_{xc} is equivalent to including $|D\rangle$ in CI.

Maitra *et al.* (2004) showed that the contribution of the doubly excited state to the exact f_{xc} takes the form:

$$f_{xc}(\omega) \sim \frac{|\langle \Phi_{ia} | V_{ee} | \Phi_D \rangle|^2}{\omega - \omega_D^{\text{KS}}} + \text{regular}, \quad (86)$$

where Φ_{ia} is the KS single-excitation Slater determinant and Φ_D is the doubly excited determinant. The numerator is the Coulomb matrix element coupling single to double excitation — exactly the CI matrix element.

Substituting Eq. (86) into the Casida coupling matrix $K_{ia,jb}(\omega)$ at $\omega \approx \omega_D^{\text{KS}}$ gives a divergence that “pulls” a new eigenvalue of Eq. (80) to the position Ω_D — reproducing the double excitation.

With the *adiabatic* kernel $f_{xc}(\omega) = \text{const}$, the pole is absent, and Ω_D never appears as an eigenvalue, regardless of the strength of the driving field.

Remark 2. Using a field with symmetry $\hat{x}^2$ or $\hat{x}\hat{y}$ (two-photon absorption) drives a different physical process: it accesses states that are two-photon-allowed, which is a distinct spectroscopic phenomenon from what the Casida equations describe. The “missing double excitations” in TDDFT are states accessible by *one photon* whose energy is wrong because of missing CI mixing.

Energy shift from CI: perturbation theory estimate. The second-order energy correction from the coupling between the singly excited state $|S\rangle$ at ω_S and the doubly excited state $|D\rangle$ at ω_D is:

$$\Delta\omega_S = \frac{|\langle S|V_{ee}|D\rangle|^2}{\omega_S - \omega_D}. \quad (87)$$

Adiabatic TDDFT misses this shift entirely. The magnitude can be 0.2–0.5 eV in conjugated molecules, explaining why the TDDFT absorption spectrum of polyenes is systematically too high in energy.

6 Gauge Invariance and Current-Density TDDFT

6.1 Gauge Transformations in Quantum Mechanics

The Hamiltonian for an electron in an electromagnetic field is (in Gaussian units converted to atomic units):

$$H = \frac{1}{2}(\hat{\mathbf{p}} + \mathbf{A}(\mathbf{r}, t))^2 + \phi(\mathbf{r}, t), \quad (88)$$

where $\mathbf{A}$ is the vector potential and ϕ is the scalar potential. The physical electric and magnetic fields are:

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (89)$$

A *gauge transformation* with a scalar function $\chi(\mathbf{r}, t)$ leaves $\mathbf{E}$ and $\mathbf{B}$ unchanged while transforming the potentials:

$$\mathbf{A} \rightarrow \mathbf{A}' = \mathbf{A} + \nabla\chi, \quad (90)$$

$$\phi \rightarrow \phi' = \phi - \frac{\partial\chi}{\partial t}. \quad (91)$$

The wavefunction transforms as:

$$\Psi(\mathbf{r}, t) \rightarrow \Psi'(\mathbf{r}, t) = e^{i\chi(\mathbf{r}, t)} \Psi(\mathbf{r}, t). \quad (92)$$

One can verify directly that Ψ' satisfies the Schrödinger equation with (H', Ψ') if and only if Ψ satisfies it with (H, Ψ) .

6.2 The Charge Density Is Gauge-Invariant

Derivation

Under Eq. (92):

$$\rho'(\mathbf{r}, t) = |\Psi'|^2 = |e^{i\chi}\Psi|^2 = e^{-i\chi}e^{+i\chi}|\Psi|^2 = |\Psi|^2 = \rho(\mathbf{r}, t). \quad (93)$$

The charge density is **gauge-invariant**: two different gauges related by χ give identical $\rho(\mathbf{r}, t)$ for all t .

6.3 The Paramagnetic Current Is NOT Gauge-Invariant

The paramagnetic (canonical) current density is defined as:

$$\mathbf{J}_p(\mathbf{r}, t) = \frac{1}{2i} [\Psi^*(\mathbf{r}, t) \nabla \Psi(\mathbf{r}, t) - \Psi(\mathbf{r}, t) \nabla \Psi^*(\mathbf{r}, t)]. \quad (94)$$

Derivation

Under the gauge transformation $\Psi \rightarrow e^{i\chi}\Psi$, compute $\nabla \Psi'$:

$$\nabla \Psi' = \nabla(e^{i\chi}\Psi) = i(\nabla\chi) e^{i\chi}\Psi + e^{i\chi}\nabla\Psi. \quad (95)$$

Substituting into Eq. (94):

$$\begin{aligned} \mathbf{J}'_p &= \frac{1}{2i} [(e^{-i\chi}\Psi^*)(i(\nabla\chi)e^{i\chi}\Psi + e^{i\chi}\nabla\Psi) - \text{c.c.}] \\ &= \frac{1}{2i} [\Psi^*\nabla\Psi - \Psi\nabla\Psi^*] + \frac{1}{2i} [i\nabla\chi|\Psi|^2 + i\nabla\chi|\Psi|^2] \\ &= \mathbf{J}_p + \rho \nabla\chi. \end{aligned} \quad (96)$$

Key Result

$\mathbf{J}_p$ is **not** gauge-invariant: it changes by $\rho \nabla\chi$ under the transformation.

6.4 The Physical (Total) Current Is Gauge-Invariant

Canonical vs kinetic momentum. In classical mechanics, a charged particle in a vector potential $\mathbf{A}$ has two distinct momenta:

- **Canonical momentum:** $\mathbf{p} = m\mathbf{v} + e\mathbf{A}$ (the quantity conjugate to position in the Lagrangian; gauge-dependent).
- **Kinetic momentum:** $m\mathbf{v} = \mathbf{p} - e\mathbf{A}$ (the physically observable momentum that equals $m \times \text{velocity}$; gauge-invariant).

The same distinction carries over to quantum mechanics. The paramagnetic current $\mathbf{J}_p$ (Section ??) is built from $\mathbf{p} = -i\nabla$ — the *canonical* momentum operator. The physical, measurable current is built from the *kinetic* momentum $(-i\nabla - e\mathbf{A})$, giving:

$$\mathbf{J}(\mathbf{r}, t) = \underbrace{\mathbf{J}_p(\mathbf{r}, t)}_{\text{canonical part}} - \underbrace{\rho(\mathbf{r}, t) \mathbf{A}(\mathbf{r}, t)}_{\text{diamagnetic part}}. \quad (97)$$

The second term $-\rho\mathbf{A}$ is called the **diamagnetic current**: it represents the response of the charge density to the vector potential itself, and it is what makes $\mathbf{J}$ gauge-invariant while $\mathbf{J}_p$ alone is not.

Derivation

Under the gauge transformation $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$, $\mathbf{J}_p \rightarrow \mathbf{J}_p + \rho\nabla\chi$:

$$\begin{aligned}\mathbf{J}' &= \mathbf{J}'_p - \rho'\mathbf{A}' \\ &= (\mathbf{J}_p + \rho\nabla\chi) - \rho(\mathbf{A} + \nabla\chi) \\ &= \mathbf{J}_p - \rho\mathbf{A} = \mathbf{J}.\end{aligned}\tag{98}$$

Key Result

$\mathbf{J}(\mathbf{r}, t) = \mathbf{J}_p - \rho\mathbf{A}$ is **gauge-invariant**. It is the physically observable current that appears in Maxwell's equations.

6.5 Why ρ Alone Fails for Periodic Systems

For an isolated finite system, the Runge-Gross theorem establishes a one-to-one mapping $V_{\text{ext}}(t) \leftrightarrow \rho(t)$ even without discussing the gauge. The reason: for a finite system, one can always choose a unique gauge by requiring, e.g., $\mathbf{A} = 0$ (the Coulomb gauge).

For a *periodic crystal*, a spatially uniform (but time-dependent) vector potential $\mathbf{A}(t)$ drives a macroscopic current. Under a gauge transformation with $\chi(\mathbf{r}, t) = -\mathbf{A}(t) \cdot \mathbf{r}$:

$$\mathbf{A}(t) \rightarrow \mathbf{A}' = \mathbf{A}(t) + \nabla(-\mathbf{A}(t) \cdot \mathbf{r}) = \mathbf{A}(t) - \mathbf{A}(t) = 0,\tag{99}$$

$$\phi \rightarrow \phi' = \phi + \dot{\mathbf{A}}(t) \cdot \mathbf{r} = \mathbf{E}(t) \cdot \mathbf{r}.\tag{100}$$

This gauge transformation relates the *velocity gauge* (vector potential $\mathbf{A}$, no scalar potential) to the *length gauge* (scalar potential $\mathbf{E} \cdot \mathbf{r}$, no vector potential).

The critical observation: in a crystal, $\chi = -\mathbf{A}(t) \cdot \mathbf{r}$ is *spatially linear*, so $\nabla\chi$ is uniform. A uniform $\nabla\chi$ does not break the periodicity of $\rho(\mathbf{r}, t)$: the density remains periodic regardless of the gauge. Therefore, **measuring $\rho(\mathbf{r}, t)$ alone cannot distinguish the velocity gauge from the length gauge** — it cannot detect whether $\mathbf{A}(t)$ is present. But $\mathbf{J}(\mathbf{r}, t)$ can: the macroscopic current changes sign when $\mathbf{A}(t)$ reverses.

This is the fundamental reason why the Runge-Gross theorem in its original form does not apply to periodic solids: the mapping $\mathbf{A}(t) \leftrightarrow \rho(t)$ is not one-to-one.

6.6 The Vignale-Kohn Theorem

Vignale and Kohn (1996) extended the Runge-Gross argument to the pair $(\rho(\mathbf{r}, t), \mathbf{J}(\mathbf{r}, t))$:

Theorem 1 (Vignale-Kohn 1996). Given a fixed initial state Ψ_0 , there is a one-to-one mapping between the physical electromagnetic field and the pair $(\rho(\mathbf{r}, t), \mathbf{J}(\mathbf{r}, t))$. Both ρ and $\mathbf{J}$ together uniquely determine all physical observables.

Since both ρ and $\mathbf{J}$ are gauge-invariant (proved above), this theorem is gauge-consistent. The basic variable of current-density TDDFT (CDFT) is therefore the pair $(\rho, \mathbf{J})$, and the XC functional is:

$$V_{\text{xc}}^{\text{CDFT}}[\rho, \mathbf{J}; \mathbf{r}, t] = V_{\text{xc}}^{\text{CDFT}}[\rho(\mathbf{r}', t'), \mathbf{J}(\mathbf{r}', t'); t' \leq t].\tag{101}$$

6.7 What Is a Kernel?

The word *kernel* appears repeatedly in this document — the XC kernel, the memory kernel, the viscosity kernel. Before applying these concepts to the electron gas, we build the mathematical intuition from the ground up.

6.7.1 Integral Operators and Their Kernels

An *integral operator* $\mathcal{K}$ maps an input function f to an output function g via integration:

$$g(x) = (\mathcal{K}f)(x) = \int K(x, y) f(y) dy. \quad (102)$$

The function $K(x, y)$ is the *kernel* of the operator. It encodes the rule: “how much does the value of f at point y contribute to the result at point x ?”

Three everyday examples.

1. **Gaussian smoothing** (image processing):

$$g(x) = \int \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-y)^2/2\sigma^2} f(y) dy. \quad (103)$$

The kernel $K(x, y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-y)^2/2\sigma^2}$ is a Gaussian centered at $y = x$. Each output value is a weighted average of nearby input values. Points far from x contribute little; points close to x contribute much.

2. **Hartree (Coulomb) potential:**

$$V_H(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'. \quad (104)$$

The kernel is $K(\mathbf{r}, \mathbf{r}') = 1/|\mathbf{r} - \mathbf{r}'|$. The Hartree potential at $\mathbf{r}$ is the sum of Coulomb contributions from the density at every other point $\mathbf{r}'$. This is a *nonlocal* operator in space: $V_H(\mathbf{r})$ depends on ρ everywhere, not just at $\mathbf{r}$.

3. **Green’s function of the Laplacian:** The Poisson equation $-\nabla^2 V = 4\pi\rho$ has the solution $V(\mathbf{r}) = \int G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d^3r'$ with $G(\mathbf{r}, \mathbf{r}') = 1/|\mathbf{r} - \mathbf{r}'|$. Green’s functions are kernels of the inverse differential operator.

6.7.2 Convolution Kernels: Translation-Invariant Operators

When the kernel depends only on the difference $K(x, y) = K(x - y)$, the integral operator is a *convolution*:

$$g(x) = \int K(x - y) f(y) dy \equiv (K * f)(x). \quad (105)$$

Convolution kernels are particularly natural for:

- Homogeneous systems (same physics at every point).
- Time-domain response (system looks the same at every time t).
- Frequency analysis: convolution in time $\leftrightarrow$ *multiplication* in frequency, $\hat{g}(\omega) = \hat{K}(\omega) \hat{f}(\omega)$.

6.7.3 Causal (Memory) Kernels

In physics, response functions must be *causal*: the effect cannot precede its cause. A causal convolution kernel satisfies:

$$K(\tau) = 0 \quad \text{for } \tau < 0, \quad (106)$$

so the response at time t depends only on inputs at times $t' < t$:

$$g(t) = \int_0^t K(t - t') f(t') dt' = \int_0^\infty K(\tau) f(t - \tau) d\tau. \quad (107)$$

The kernel $K(\tau)$ tells you “how strongly does an input from τ seconds ago influence the current output?” If $K(\tau) \propto e^{-\tau/\tau_0}$, the influence decays exponentially with a characteristic memory time τ_0 .

Physical Insight

A system with a causal memory kernel is the mathematical formalization of your car analogy. The kernel $K(\tau)$ weights the past trajectory $f(t')$ of the input — car position, density, deformation — and the current output $g(t)$ depends on the full weighted history. The adiabatic approximation corresponds to $K(\tau) = c\delta(\tau)$: zero memory, only the instantaneous value matters.

6.7.4 The XC Kernel in DFT

The *exchange-correlation kernel* f_{xc} is the functional derivative of the XC potential with respect to the density:

$$f_{xc}(\mathbf{r}, \mathbf{r}', t - t') = \frac{\delta V_{xc}(\mathbf{r}, t)}{\delta \rho(\mathbf{r}', t')}. \quad (108)$$

This answers the question: “if I add a tiny density perturbation $\delta\rho(\mathbf{r}')$ at time t' , how does $V_{xc}(\mathbf{r})$ respond at a later time t ?”

In the *static* (ground-state) limit, the XC kernel reduces to the second functional derivative of the XC energy:

$$f_{xc}^{gs}(\mathbf{r}, \mathbf{r}') = \frac{\delta^2 E_{xc}[\rho]}{\delta \rho(\mathbf{r}) \delta \rho(\mathbf{r}')}. \quad (109)$$

For LDA, where $E_{xc} = \int \epsilon_{xc}(n) \rho d^3r$:

$$f_{xc}^{LDA}(\mathbf{r}, \mathbf{r}') = \frac{d^2[\rho \epsilon_{xc}(\rho)]}{d\rho^2} \delta(\mathbf{r} - \mathbf{r}'). \quad (110)$$

The delta function $\delta(\mathbf{r} - \mathbf{r}')$ makes LDA *local* in space: the XC potential at $\mathbf{r}$ depends only on the density at the same point $\mathbf{r}$. GGA kernels have a spatial gradient correction (nonzero for $|\mathbf{r} - \mathbf{r}'| \lesssim$ grid spacing). The frequency-dependent kernel $f_{xc}(\omega)$ additionally captures nonlocality *in time* — the memory.

6.8 Elements of Continuum Mechanics

The Vignale-Kohn functional treats the electron gas as a *viscoelastic fluid* — a material that has both elastic (spring-like) and viscous (dashpot-like) properties, with a memory of its deformation history. This section builds the necessary continuum mechanics background from scratch.

6.8.1 Displacement Field and Velocity Field

Consider a fluid (or solid) in which each material element at position $\mathbf{r}_0$ (its reference position) is displaced to a new position $\mathbf{r}_0 + \mathbf{u}(\mathbf{r}_0, t)$ at time t . The vector field $\mathbf{u}(\mathbf{r}, t)$ is the *displacement field*.

The *velocity field* of the fluid elements is:

$$\mathbf{v}(\mathbf{r}, t) = \frac{\partial \mathbf{u}(\mathbf{r}, t)}{\partial t}. \quad (111)$$

For the electron gas, $\mathbf{u}(\mathbf{r}, t)$ is the displacement of the electron fluid element at $\mathbf{r}$, and the current density is:

$$\mathbf{J}(\mathbf{r}, t) = n_0(\mathbf{r}) \mathbf{v}(\mathbf{r}, t) = n_0(\mathbf{r}) \frac{\partial \mathbf{u}}{\partial t}, \quad (112)$$

where $n_0(\mathbf{r})$ is the equilibrium electron density.

6.8.2 The Strain Tensor: Measuring Deformation

The *strain tensor* $\varepsilon_{\alpha\beta}$ measures how much the material is locally deformed:

$$\varepsilon_{\alpha\beta} = \frac{1}{2} \left(\frac{\partial u_\alpha}{\partial r_\beta} + \frac{\partial u_\beta}{\partial r_\alpha} \right). \quad (113)$$

It is symmetric ($\varepsilon_{\alpha\beta} = \varepsilon_{\beta\alpha}$) and captures two types of deformation:

Volumetric (compressive) part — the trace. The trace of the strain tensor measures volume change:

$$\varepsilon_{\text{vol}} = \text{Tr}(\varepsilon) = \nabla \cdot \mathbf{u} = \frac{\delta V}{V_0}. \quad (114)$$

$\nabla \cdot \mathbf{u} > 0$: expansion; $\nabla \cdot \mathbf{u} < 0$: compression.

Deviatoric (shear) part — traceless. The traceless part of ε describes shape change without volume change:

$$e_{\alpha\beta} = \varepsilon_{\alpha\beta} - \frac{1}{3}(\nabla \cdot \mathbf{u}) \delta_{\alpha\beta}. \quad (115)$$

This is the *shear strain*: it measures how the angles between material lines change.

Notation: Three Roles of $e_{\alpha\beta}$

The symbol $e_{\alpha\beta}$ carries three related but distinct meanings in this section.

Symbol	Object	Equation
$\varepsilon_{\alpha\beta}$	full strain tensor (symmetric)	(113)
$e_{\alpha\beta} = \varepsilon_{\alpha\beta} - \frac{1}{3}(\nabla \cdot \mathbf{u})\delta_{\alpha\beta}$	deviatoric (traceless) strain	(115)
$\dot{e}_{\alpha\beta} \equiv \partial e_{\alpha\beta} / \partial t$	deviatoric strain <i>rate</i>	(120)

In the VK functional (Eq. 122) the relevant quantity is the deviatoric strain *rate* $\dot{e}_{\alpha\beta}$ (time derivative of the traceless part), sometimes written $e^{\alpha\beta}$ in index-raising notation. When reading the VK equation, $e^{\alpha\beta}$ always means the strain rate, not the strain itself.

Derivation

Concrete example: simple shear. Suppose the displacement is purely in the x -direction and varies only with y : $\mathbf{u} = (u_x(y), 0, 0)$. Then:

$$\varepsilon_{xy} = \varepsilon_{yx} = \frac{1}{2} \frac{\partial u_x}{\partial y}, \quad \text{all other components zero.}$$

The material is being sheared: layers at different y slide past each other. A square element deforms into a parallelogram. No volume change ($\nabla \cdot \mathbf{u} = 0$), only angular distortion.

In the VK functional, the relevant quantity is the *strain rate tensor* — the time derivative of the strain:

$$\dot{e}_{\alpha\beta}(\mathbf{r}, t) = \frac{\partial e_{\alpha\beta}}{\partial t} = \frac{1}{2} \left(\frac{\partial v_\alpha}{\partial r_\beta} + \frac{\partial v_\beta}{\partial r_\alpha} \right) - \frac{1}{3} (\nabla \cdot \mathbf{v}) \delta_{\alpha\beta}, \quad (116)$$

where $\mathbf{v} = \partial \mathbf{u} / \partial t$ is the fluid velocity.

6.8.3 The Stress Tensor: Internal Forces

When a material is deformed, internal restoring forces develop. The *stress tensor* $\sigma_{\alpha\beta}$ describes these forces: $\sigma_{\alpha\beta}$ is the force per unit area in the α -direction acting on a surface whose outward normal points in the β -direction.

- **Diagonal elements** $\sigma_{\alpha\alpha}$: normal stresses (tension or compression along α).
- **Off-diagonal elements** $\sigma_{\alpha\beta}$ ($\alpha \neq \beta$): shear stresses (forces parallel to the surface).
- **Symmetry**: $\sigma_{\alpha\beta} = \sigma_{\beta\alpha}$ (required so that fluid elements have no net torque).

The equation of motion for a fluid element is Newton's second law for a continuum:

$$\rho_m \frac{\partial v_\alpha}{\partial t} = \sum_\beta \frac{\partial \sigma_{\alpha\beta}}{\partial r_\beta} + f_\alpha, \quad (117)$$

where ρ_m is the mass density and f_α is any external body force per unit volume. For the electron gas, this becomes the force balance between the XC stress tensor and the XC electric field (Eq. 121).

6.8.4 Constitutive Relations: How Stress Relates to Strain

A *constitutive relation* specifies how the stress $\sigma_{\alpha\beta}$ depends on the deformation. Different materials obey different constitutive relations:

1. Elastic solid (Hooke’s law). Stress is proportional to the instantaneous strain:

$$\sigma_{\alpha\beta} = G_0 e_{\alpha\beta} + K_0 (\nabla \cdot \mathbf{u}) \delta_{\alpha\beta}, \quad (118)$$

where G_0 is the shear modulus and K_0 is the bulk modulus. No memory: the stress depends only on the current deformation. If the force is removed, the material springs back instantly (rubber on short timescales).

2. Viscous Newtonian fluid (Newton’s law of viscosity). Stress is proportional to the instantaneous strain *rate*:

$$\sigma_{\alpha\beta}^{\text{visc}} = \eta \dot{e}_{\alpha\beta} + \zeta (\nabla \cdot \mathbf{v}) \delta_{\alpha\beta}, \quad (119)$$

where η is the *shear viscosity* (resistance to sliding) and ζ is the *bulk viscosity* (resistance to compression). Again no memory: only the current rate of deformation matters. Water is well-described by Eq. (119) at low flow speeds.

3. Viscoelastic material: memory in the constitutive relation. A viscoelastic material has both elastic and viscous character. The general constitutive relation replaces the instantaneous G_0 by a *time-convolution with a relaxation kernel* $G(\tau)$:

$$\sigma_{\alpha\beta}(t) = \int_0^\infty G(\tau) \dot{e}_{\alpha\beta}(t - \tau) d\tau = \int_{-\infty}^t G(t - t') \dot{e}_{\alpha\beta}(t') dt'. \quad (120)$$

The *relaxation modulus* $G(\tau)$ is the kernel. It tells you how strongly the shear strain rate from τ seconds ago still influences the current stress.

Derivation

Recovering elastic and viscous limits from the general form.

Elastic limit: $G(\tau) = G_0$ (constant, infinite memory). The material “remembers” its entire deformation history with equal weight. Substituting into Eq. (120):

$$\sigma(t) = G_0 \int_0^\infty \dot{e}(t - \tau) d\tau = G_0 [e(t) - e(-\infty)] = G_0 e(t),$$

where the last step uses $e(-\infty) = 0$ (the solid starts undeformed). This is stress proportional to strain — Hooke’s law. ✓

Viscous limit: $G(\tau) = \eta \delta(\tau)$ (instantaneous memory, no history). Substituting:

$$\sigma(t) = \int_0^\infty \eta \delta(\tau) \dot{e}(t - \tau) d\tau = \eta \dot{e}(t).$$

Stress proportional to strain *rate* — Newton’s law of viscosity. ✓

General viscoelastic: $G(\tau)$ is a smoothly decaying function, e.g. $G(\tau) = G_\infty + (G_0 - G_\infty)e^{-\tau/\tau_{\text{rel}}}$.

- For $\omega\tau_{\text{rel}} \gg 1$ (fast deformation): the material has no time to relax — behaves like an elastic solid.
- For $\omega\tau_{\text{rel}} \ll 1$ (slow deformation): the material relaxes before the next cycle — behaves like a viscous fluid.

Silly putty, polymer melts, and — crucially — the electron gas all behave this way.

6.8.5 Summary: Mechanical Analogies for the VK Functional

The Vignale-Kohn XC stress tensor (Eq. 122) is Eq. (120) applied to the electron gas:

- $e_{\alpha\beta}(t') \rightarrow$ strain rate of the electron fluid at past time t' , computed from the current density $\mathbf{J}$ (Eq. 112).
- $G(\tau) \rightarrow \eta_{\text{xc}}(n_0, \tau)$ and $\zeta_{\text{xc}}(n_0, \tau)$, the XC shear and bulk viscosity kernels of the homogeneous electron gas.
- $\sigma_{\alpha\beta}(t) \rightarrow \sigma_{\text{xc}}^{\alpha\beta}(\mathbf{r}, t)$, the XC stress that drives the correction to the KS vector potential.

The kernels η_{xc} and ζ_{xc} are material properties of the electron gas — analogous to the viscosity of water — that must be computed from quantum Monte Carlo for the homogeneous electron gas and then applied locally via the LDA of CDFT.

6.9 The Electron Fluid as a Viscoelastic Medium

Vignale and Kohn showed that for a slowly varying perturbation (the “gradient” approximation in space), the XC contribution can be expressed as a *stress tensor* $\overleftrightarrow{\sigma}_{\text{xc}}$, relating to the XC vector potential via:

$$n_0(\mathbf{r}) \mathbf{E}_{\text{xc}}(\mathbf{r}, t) = \nabla \cdot \overleftrightarrow{\sigma}_{\text{xc}}(\mathbf{r}, t), \quad (121)$$

where $\mathbf{E}_{\text{xc}} = -\partial \mathbf{A}_{\text{xc}} / \partial t$.

Connection to your car analogy. This is the point where the analogy becomes exact. The stress tensor has the form of a viscoelastic constitutive relation:

$$\sigma_{\text{xc}}^{\alpha\beta}(\mathbf{r}, t) = \int_{-\infty}^t \underbrace{\eta_{\text{xc}}(n_0, t - t')}_{\text{shear viscosity kernel}} \underbrace{e^{\alpha\beta}(\mathbf{r}, t')}_{\text{strain rate}} dt' + \int_{-\infty}^t \underbrace{\zeta_{\text{xc}}(n_0, t - t')}_{\text{bulk viscosity kernel}} \underbrace{(\nabla \cdot \mathbf{u}) \delta^{\alpha\beta}}_{\text{compression}} dt', \quad (122)$$

where:

- $\mathbf{u}(\mathbf{r}, t)$ is the **displacement field** of the electron fluid, related to the current by $\mathbf{J}(\mathbf{r}, t) = n_0(\mathbf{r}) \partial \mathbf{u} / \partial t$;
- $e^{\alpha\beta} = \frac{1}{2}(\partial_\alpha u_\beta + \partial_\beta u_\alpha) - \frac{1}{3}\delta_{\alpha\beta} \nabla \cdot \mathbf{u}$ is the **symmetrized strain rate tensor** (shear deformation);
- $\eta_{xc}(n_0, \tau)$ and $\zeta_{xc}(n_0, \tau)$ are the **shear and bulk viscosity kernels** of the homogeneous electron gas at density n_0 , computable from quantum Monte Carlo.

Physical Insight

The strain rate $e^{\alpha\beta}$ encodes how fast the electron fluid is being sheared at time t' : it is exactly your car analogy. The XC response today (stress σ_{xc} at time t) depends on the full history of the deformation $e^{\alpha\beta}(\mathbf{r}, t')$ for all past times $t' \leq t$, weighted by the viscosity kernel. A static electron cloud produces zero shear and zero VK correction; a rapidly oscillating current produces a large strain-rate history and a large VK correction.

6.10 Viscosity Kernels: Explicit Forms and Limits

The kernels are related to the frequency-dependent XC moduli of the homogeneous electron gas:

$$\eta_{xc}(n_0, \omega) = -\frac{n_0^2}{\omega} \text{Im } f_{xc}^T(n_0, \omega), \quad (123)$$

$$\zeta_{xc}(n_0, \omega) = -\frac{n_0^2}{\omega} [\text{Im } f_{xc}^L(n_0, \omega) - \frac{4}{3} \text{Im } f_{xc}^T(n_0, \omega)], \quad (124)$$

where f_{xc}^T and f_{xc}^L are the **transverse** and **longitudinal** XC kernels of the uniform electron gas. These labels refer to the direction of the density fluctuation relative to its wavevector $\mathbf{q}$:

- **Longitudinal** (f_{xc}^L): the density fluctuation is *parallel* to $\mathbf{q}$ — a compressional (sound-wave) mode. This is the only component that survives in ordinary TDDFT based on the density ρ alone, because ρ is a scalar (longitudinal field).
- **Transverse** (f_{xc}^T): the density fluctuation is *perpendicular* to $\mathbf{q}$ — a shear mode (like vorticity in a fluid). Transverse response requires the vector current density $\mathbf{J}$ as the basic variable, which is why current-DFT (VK) is needed: ordinary TDDFT has $f_{xc}^T = 0$ by construction.

In the $q \rightarrow 0$ (long-wavelength) limit relevant to optical response, both f_{xc}^L and f_{xc}^T approach finite values that are the XC compressional and shear moduli of the electron gas.

The known limiting behaviors are:

$$\lim_{\omega \rightarrow 0} \text{Im } f_{xc}^T(\omega) \propto \omega \quad (\text{Drude-like, weak low-frequency dissipation}), \quad (125)$$

$$\lim_{\omega \rightarrow \infty} \text{Im } f_{xc}^T(\omega) \propto \omega^{-3/2} \quad (\text{high-frequency algebraic decay}). \quad (126)$$

In the time domain, $\eta_{\text{xc}}(\tau)$ is a causal function that decays over a characteristic time $\sim 1/\omega_p$ (the inverse plasma frequency), meaning the memory has a finite range. This is the basis for the practical truncation of the history integral in numerical implementations.

7 Functional Forms and Implementation of Memory Functionals

7.1 The Simplest Memory Extension: Linear Convolution on Top of LDA

The minimal memory extension retains the scalar V_{xc} form and adds a linear convolution:

$$V_{\text{xc}}(\mathbf{r}, t) = V_{\text{xc}}^{\text{LDA}}[\rho(\mathbf{r}, t)] + \int_0^t K_{\text{xc}}(n_0(\mathbf{r}), t - t') \delta\rho(\mathbf{r}, t') dt', \quad (127)$$

where $\delta\rho = \rho(t') - \rho_{\text{gnd}}$ and $K_{\text{xc}}(n_0, \tau)$ is the Fourier transform of $[f_{\text{xc}}(n_0, \omega) - f_{\text{xc}}(n_0, 0)]$ — the frequency-dependent correction relative to the static (adiabatic) value.

This form:

- Reduces to adiabatic LDA when $K_{\text{xc}} = 0$ (or $\omega \rightarrow 0$).
- Captures shifts in excitation energies due to frequency-dependent screening.
- Does not capture full VK dissipation (which requires the vector potential $\mathbf{A}_{\text{xc}}$).

Implementation cost.

1. Store $\delta\rho(\mathbf{r}, t')$ for $t' \in [t - \tau_{\text{max}}, t]$: array of size $N_{\text{grid}} \times N_{\text{history}}$.
2. At each step: convolve with $K_{\text{xc}}(n_0(\mathbf{r}), \tau)$: $O(N_{\text{grid}} \times N_{\text{history}})$ operations.
3. Truncate history at $\tau_{\text{max}} \sim 10/\omega_p$ (kernel is exponentially small beyond this).
4. Memory overhead: $\sim N_{\text{grid}} \times 100$ doubles (modest).

7.2 The Vignale-Kohn Functional: Full Implementation Requirements

The full VK functional requires computing $\mathbf{A}_{\text{xc}}$ from the stress tensor via Eq. (121):

$$-n_0 \frac{\partial \mathbf{A}_{\text{xc}}}{\partial t} = \nabla \cdot \overleftrightarrow{\sigma}_{\text{xc}}. \quad (128)$$

Integrating in time gives $\mathbf{A}_{\text{xc}}(\mathbf{r}, t)$, which then enters the KS Hamiltonian as an additional velocity-gauge coupling:

$$H_{\text{KS}}(t) \rightarrow H_{\text{KS}} + [\mathbf{A}_{\text{ext}}(t) + \mathbf{A}_{\text{xc}}(\mathbf{r}, t)] \cdot \hat{\mathbf{p}}. \quad (129)$$

The matrix element of the XC correction in the KS basis is:

$$\langle \varphi_i | \mathbf{A}_{\text{xc}} \cdot \hat{\mathbf{p}} | \varphi_j \rangle = \int \mathbf{A}_{\text{xc}}(\mathbf{r}, t) \cdot \varphi_i^*(\mathbf{r}) (-i\nabla) \varphi_j(\mathbf{r}) d^3r. \quad (130)$$

This has the same structure as the external-field matrix element $P_{ij}^\alpha = iv_{\text{cell}} \langle \varphi_i | \partial_\alpha | \varphi_j \rangle$ already computed in the RMG velocity-gauge framework. The VK correction enters the same code path — it is an additional contribution to the effective $\mathbf{A}(t)$ at each timestep.

Table 1: Comparison of XC approximations for RT-TDDFT.

Functional	Memory	Dissipation	Doubles	Cost
Adiabatic LDA/GGA (current)	No	No	No	$O(N^3)$
Scalar memory [Eq. (127)]	Partial	No	Partial	$+O(N_h)$
Vignale-Kohn (VK)	Yes (vector)	Yes	Partial	$+O(N_h) + \text{Poisson}$
Exact TDDFT	Full	Full	Full	Exponential

7.3 Causality and Kramers-Kronig Relations

Any physically correct memory kernel must satisfy the *Kramers-Kronig (KK) relations*. These are not independent constraints imposed from outside — they follow automatically from causality via **complex analysis**.

Origin from causality and analyticity. A causal kernel $f_{xc}(\tau) = 0$ for $\tau < 0$ has a one-sided Fourier transform (Eq. 55). The key consequence is that $f_{xc}(\omega)$, extended to complex ω , is *analytic in the upper half-plane* $\text{Im } \omega > 0$: no poles or branch cuts there, because the $e^{i\omega\tau}$ factor is exponentially damped when $\text{Im } \omega > 0$ and $\tau > 0$.

Applying the **Cauchy integral theorem** to a contour that runs along the real axis and closes in the upper half-plane (where $f_{xc} \rightarrow 0$ at large $|\omega|$), one obtains the **dispersion relation**:

$$f_{xc}(\omega) = \frac{1}{\pi i} \mathcal{P} \int_{-\infty}^{\infty} \frac{f_{xc}(\omega')}{\omega' - \omega} d\omega', \quad (131)$$

where $\mathcal{P}$ denotes the Cauchy principal value. Separating real and imaginary parts yields the Kramers-Kronig relations ($K_{xc}(\tau) = 0$ for $\tau < 0$):

$$\text{Re } f_{xc}(\omega) = f_{xc}(0) + \frac{2}{\pi} \mathcal{P} \int_0^{\infty} \frac{\omega' \text{Im } f_{xc}(\omega')}{\omega'^2 - \omega^2} d\omega', \quad (132)$$

$$\text{Im } f_{xc}(\omega) = -\frac{2\omega}{\pi} \mathcal{P} \int_0^{\infty} \frac{\text{Re } f_{xc}(\omega') - f_{xc}(0)}{\omega'^2 - \omega^2} d\omega'. \quad (133)$$

The adiabatic approximation satisfies the $\omega \rightarrow 0$ limit of Eq. (132) trivially — it is just the static kernel. But it violates Eq. (133) because $\text{Im } f_{xc}^{\text{ATDDFT}} = 0$ for all $\omega > 0$, while the right-hand side of Eq. (133) is generally nonzero. This is another way of seeing that the adiabatic approximation is inconsistent with a causal memory.

8 Summary

The four questions addressed in this tutorial are resolved as follows:

1. What does the adiabatic approximation discard? It replaces the history-dependent XC kernel $f_{xc}(\omega)$ — a function with poles and a nonzero imaginary part — by a frequency-independent constant. This is equivalent to assuming that the XC response is instantaneous (zero memory time). Time-derivative corrections improve this approximation perturbatively but cannot reproduce poles.

2. Why is there no dissipation in adiabatic TDDFT? The fundamental reason is that the KS Hamiltonian is Hermitian and the Magnus propagator is unitary, conserving the eigenvalues of $P(t)$ (occupation numbers) exactly. Population relaxation (T_1) requires a bath and is absent regardless of f_{xc} . Pure dephasing (T_2) requires $\text{Im } f_{xc}(\omega) \neq 0$, which the adiabatic approximation sets to zero. Both effects are correctly attributed to unitarity and the absence of memory, not just one or the other.

3. Why are double excitations missing? Pure doubly excited states are dipole-dark: the matrix element $\langle EE|\hat{x}|GG\rangle = 0$ exactly (Eq. 75). In the interacting system, Coulomb mixing creates states with mixed single+double character that are dipole-accessible. Adiabatic TDDFT places these states at wrong energies because the Casida basis spans only single KS excitations. Recovering the correct energy requires a pole in f_{xc} at the double-excitation frequency (Eq. 86).

4. Why is $(\rho, \mathbf{J})$ needed for periodic systems? $\rho(\mathbf{r}, t)$ is gauge-invariant (Eq. 93) and cannot distinguish the presence or absence of a macroscopic vector potential $\mathbf{A}(t)$ in a crystal. $\mathbf{J}(\mathbf{r}, t)$ is also gauge-invariant (Eq. 98) and carries the missing information. The Vignale-Kohn theorem establishes $(\rho, \mathbf{J})$ as the correct basic variable pair. The resulting XC functional has a viscoelastic memory structure (Eq. 122) that directly formalizes the intuition that the electron system's response today depends on the full history of its deformation.

References